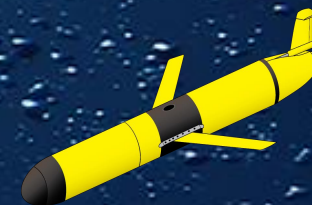
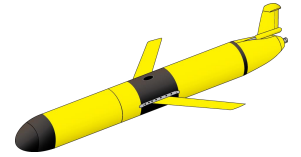




# Teledyne Webb Research Slocum Glider Training



# Slocum Glider – *The Basics*



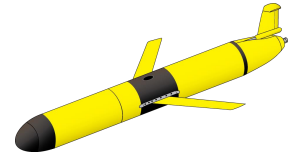
- Introductions
- Agenda for the week
- Glider Overview
  - Glider Components
  - Glider Configurations
  - Glider Flight Fundamentals
  - Glider Sensors
  - Product Applications



Photo: Tiffany Wardman



# The Slocum Glider



## What is the Slocum Glider?

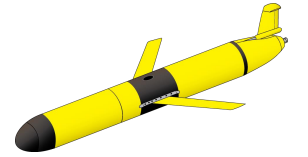
Buoyancy engine driven, ultra long-endurance AUVs for persistent ocean monitoring

## What problem does it solve?

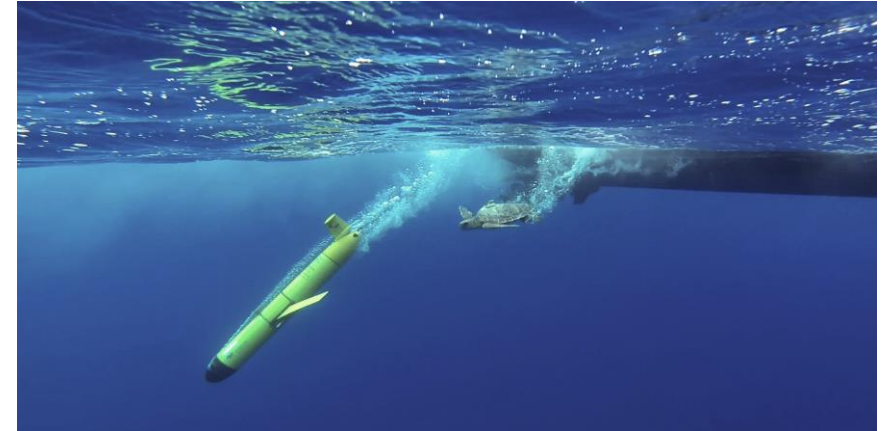
- Persistent data collection at a fraction of the cost of surface ships or traditional AUVs
- Increased data points at a higher frequency than possible with traditional means.
- Less life cycle cost
- Safer, more cost effective and higher quality data than gathered through traditional means.



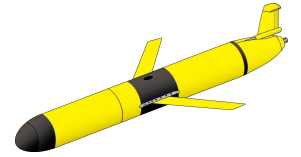
# Slocum Glider History



- Douglas Webb founded Webb Research in 1982 and developed the original Slocum Glider – named for Joshua Slocum, the first man to circumnavigate the globe solo in a sailboat
- Teledyne acquired Webb Research in 2008
- Slocum is the market leading glider – over 1000 delivered to Research, Academic, Military, and Commercial customers
- Proven performance:
  - In 2016-2020, Glider “Silbo” completed Atlantic Ocean circumnavigation with over 1,200 days at sea



# Slocum Glider History



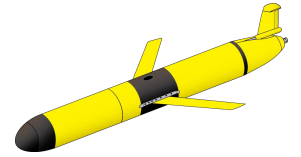
Doug Webb has a vision

**Work Hard,  
Have Fun,  
Change the World**



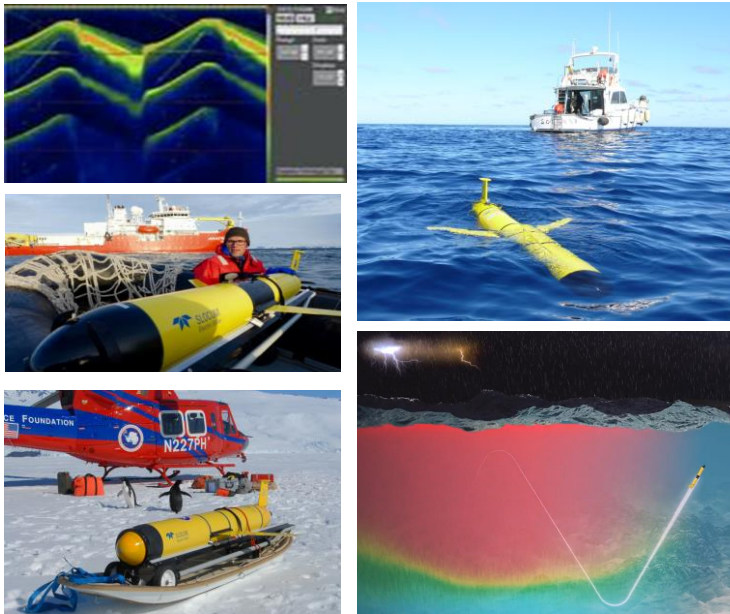


# Slocum Product Applications



## Science

- Oceanography
- Data gateway
- Ocean acidity
- Storm Survey



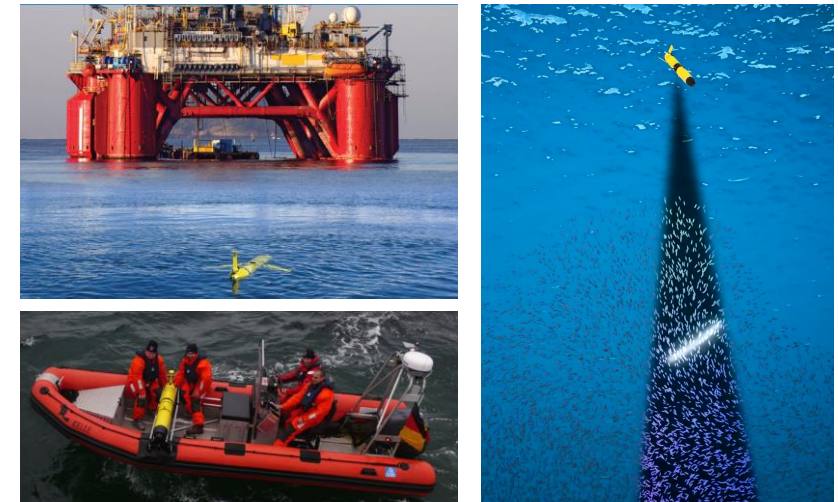
## Defense

- Oceanographic data
- Specialty builds

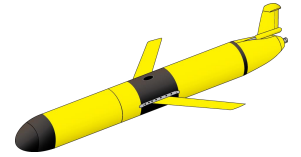


## Commercial

- Oil and Gas
- Offshore Wind
- Environmental monitoring including Current profiling
- Marine mammal mitigation
- Fisheries



# Glider Support - *Philosophy & Structure*

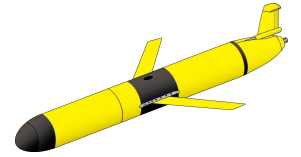


[glidersupport@teledyne.com](mailto:glidersupport@teledyne.com)

- Wide distribution list including support, service, sales, engineering, and executive personnel
- Support ticketing system
- Often will be answered during off-hours if subject is compelling enough and if personnel are available
- The “Batphone” (for first deployment “emergencies only”)
  - (508) 524-8106
  - (855) 720-3915 (toll-free - US)



# Glider Forum – *A great resource!*

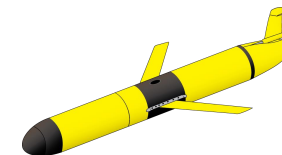


<https://datahost.webbresearch.com/>

- Most of the information we will discuss during this training is available on the Datahost, our Glider User Forum
- On the forum, you will be able to access:
  - Posts from fellow users & TWR employees on the use of gliders
  - Manuals
  - Software Releases
  - Other Glider Resources
  - Client Tools/SFMC
- Register if you have not already!
  - Once registered, contact [Cordielyn.Goodrich@teledyne.com](mailto:Cordielyn.Goodrich@teledyne.com) to have your account approved



# Glider Forum – A great resource!



## TWR Forum

Slocum Glider support and ideas

Search...   
Advanced search

[Board index](#)

[User Control Panel](#) (0 new messages) • [View your posts](#)

[FAQ](#) [Members](#) [Logout](#) [ cgoodrich ]

It is currently September 5th, 2021, 7:00 am

Last visit was: September 1st, 2021, 6:30 am

[View unanswered posts](#) • [View unread posts](#) • [View new posts](#) • [View active topics](#)

[Mark forums read](#)

| GLIDERS                                                                                                                                                                               | TOPICS | POSTS | LAST POST                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------|
|  <b>G3S New SM32 Processor</b><br>The Persistor replacement                                          | 5      | 6     | by <b>admin</b> <br>March 1st, 2021, 1:32 pm    |
|  <b>FAQ</b><br>Look here first.                                                                      | 94     | 240   | by <b>admin</b> <br>August 10th, 2021, 5:37 am  |
|  <b>Operation</b><br>Things to do with a glider in a boat                                            | 66     | 201   | by stats <br>August 10th, 2021, 3:55 pm         |
|  <b>Missions</b><br>Mission development                                                              | 65     | 187   | by Jude_Vdm <br>July 26th, 2021, 11:47 am       |
|  <b>lab testing</b>                                                                                  | 44     | 114   | by <b>admin</b> <br>July 22nd, 2021, 6:13 am    |
|  <b>Suggestions</b>                                                                                  | 28     | 107   | by ktikka <br>May 8th, 2019, 10:26 pm           |
|  <b>Aborts</b>                                                                                      | 17     | 54    | by Jude_Vdm <br>July 28th, 2021, 10:49 am       |
|  <b>NEWS!!!</b><br>Teledyne Webb Research news and share stories of your programs accomplishments! | 13     | 15    | by <b>admin</b> <br>March 17th, 2021, 6:24 am |
|  <b>Dockserver</b><br>post all GMC questions here                                                  | 92     | 305   | by <b>admin</b> <br>July 23rd, 2021, 5:02 am  |
|  <b>Resources</b><br>Links to online glider resources                                              | 2      | 2     | by <b>admin</b> <br>May 2nd, 2014, 11:15 am   |
|  <b>Glider Service Bulletins</b>                                                                   | 1      | 1     | by <b>admin</b> <br>April 9th, 2014, 10:27 am |

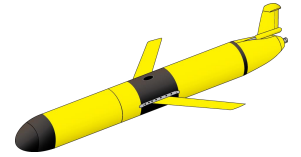
FAL-TG-21-004

EAR99-Controlled Technology Subject to Restrictions Contained on the Cover Page



**TELEDYNE MARINE**  
Everywhere you look™

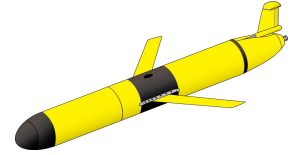
# Objectives of Glider Training



- A trainee should be comfortable configuring & preparing a glider for a qualification or test flight
- Answer glider prep, maintenance, and piloting questions for all users
- We recommend starting slow and building a base of comfort and knowledge



# Glider Basics



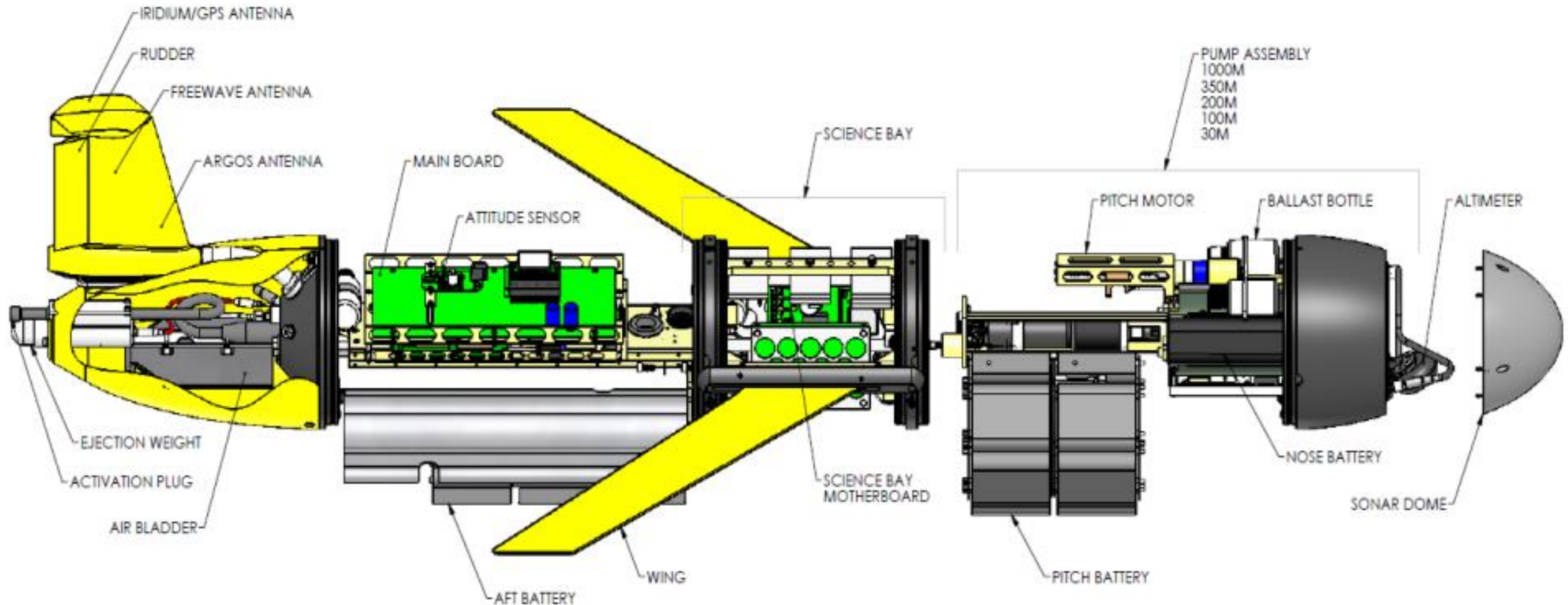
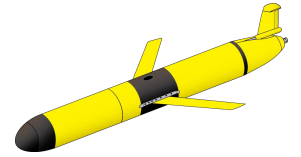
- Glider components
- Glider flight fundamentals
- Glider communication
- Shore side software - SFMC
- On-board glider software hierarchy



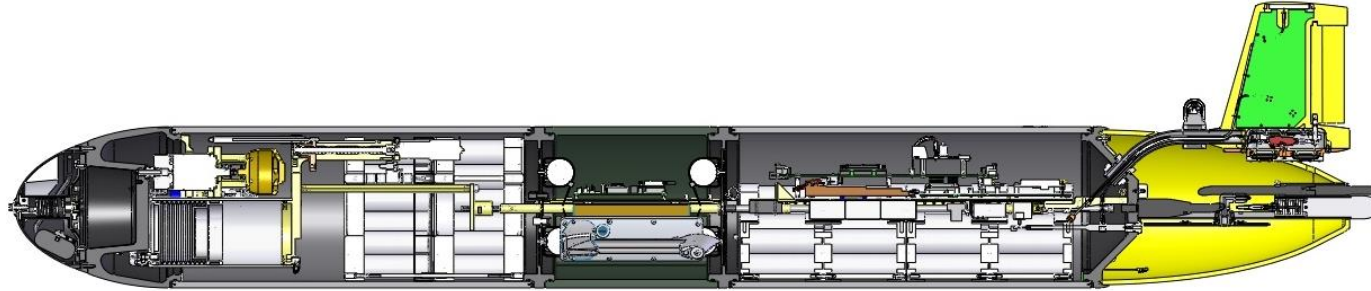
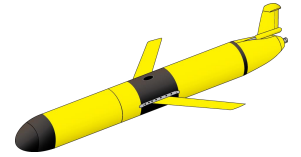
Photo: Tiffany Wardman



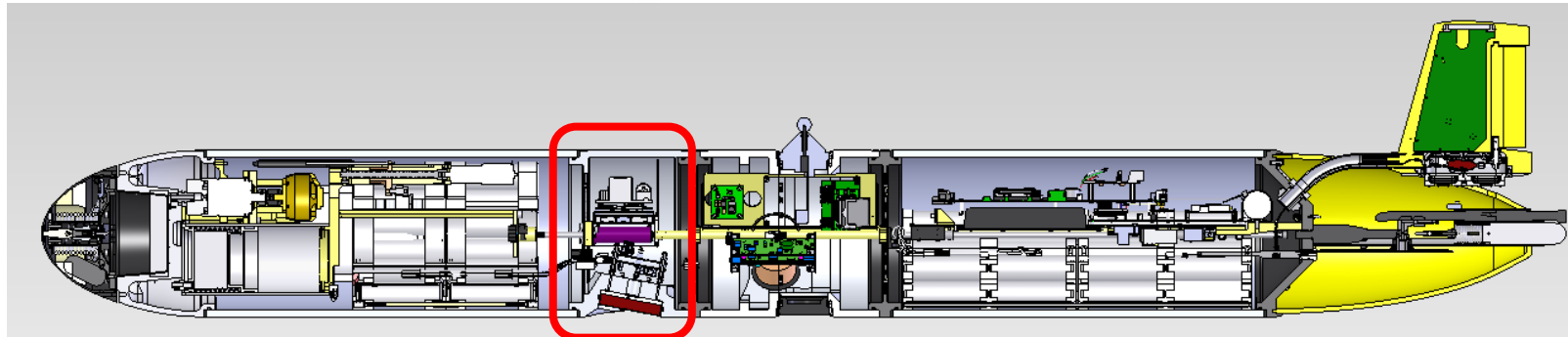
# Slocum Glider Components



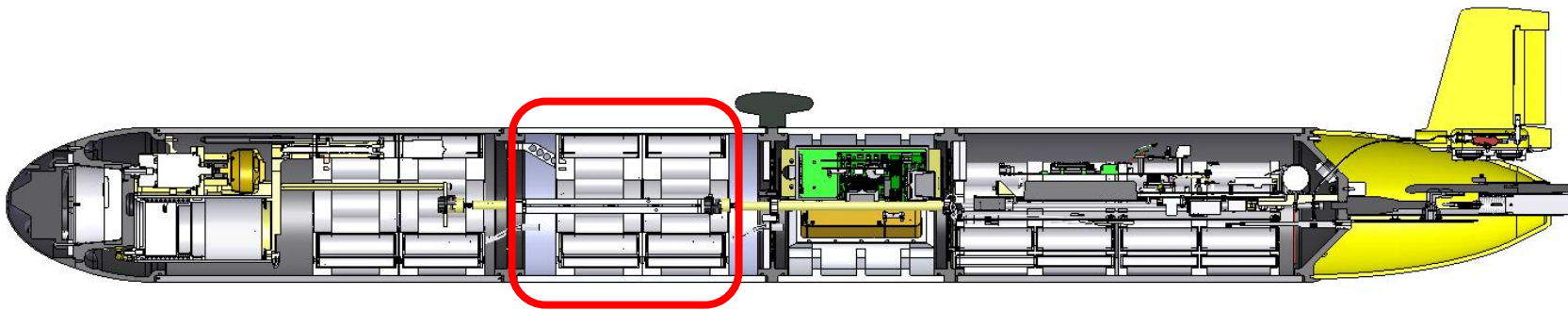
# Slocum Glider Configurations



**Standard Length Glider**

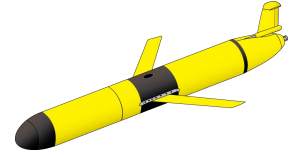


**Extended Glider with  
Additional Sensor Bay**



**Extended Glider with  
Extended Endurance Bay**

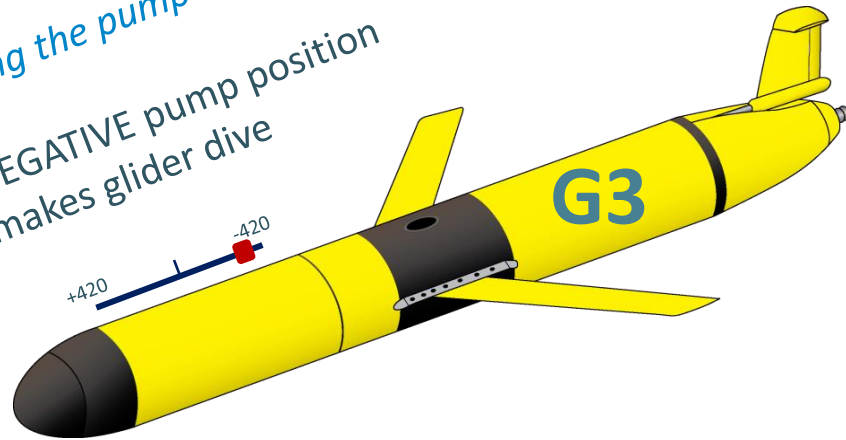
# Buoyancy Pump Positions



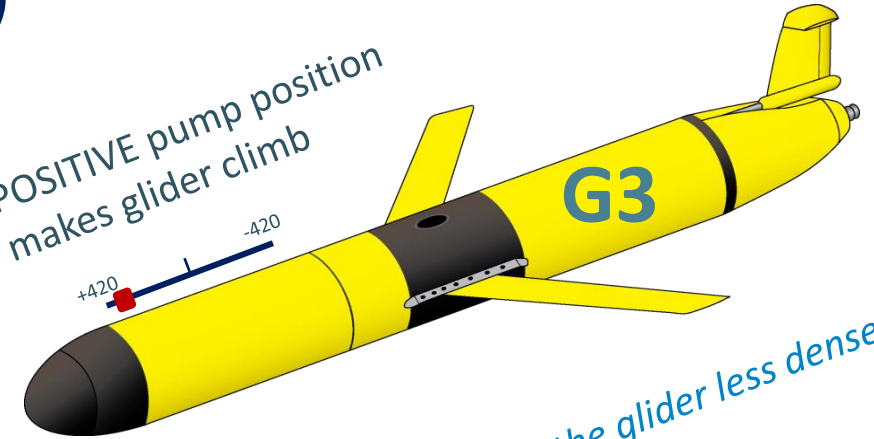
Changing the position of the pump changes the volume of the vehicle (Pump position is measured in cc's)

*Retracting the pump makes the glider more dense*

*NEGATIVE pump position  
makes glider dive*



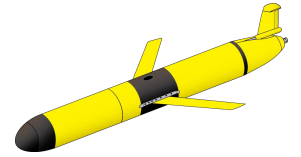
*POSITIVE pump position  
makes glider climb*



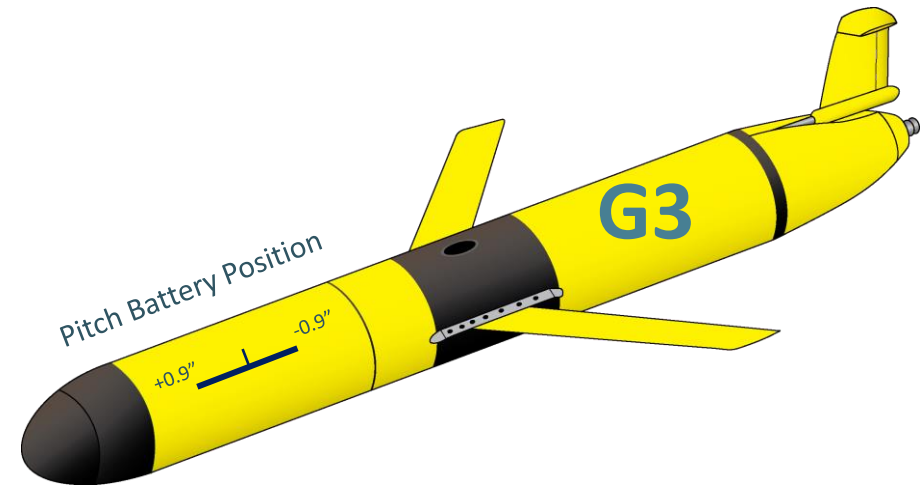
*Extending the pump makes the glider less dense*



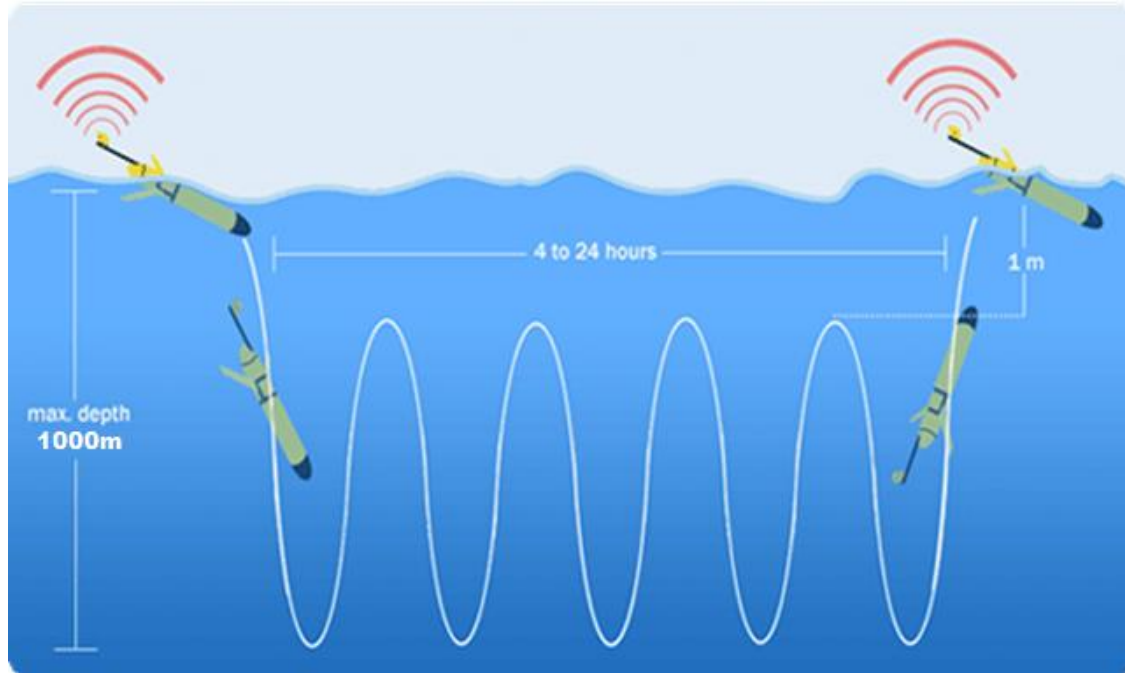
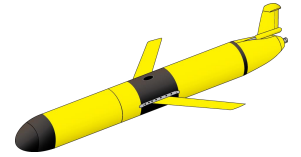
# Pitch Battery Positions



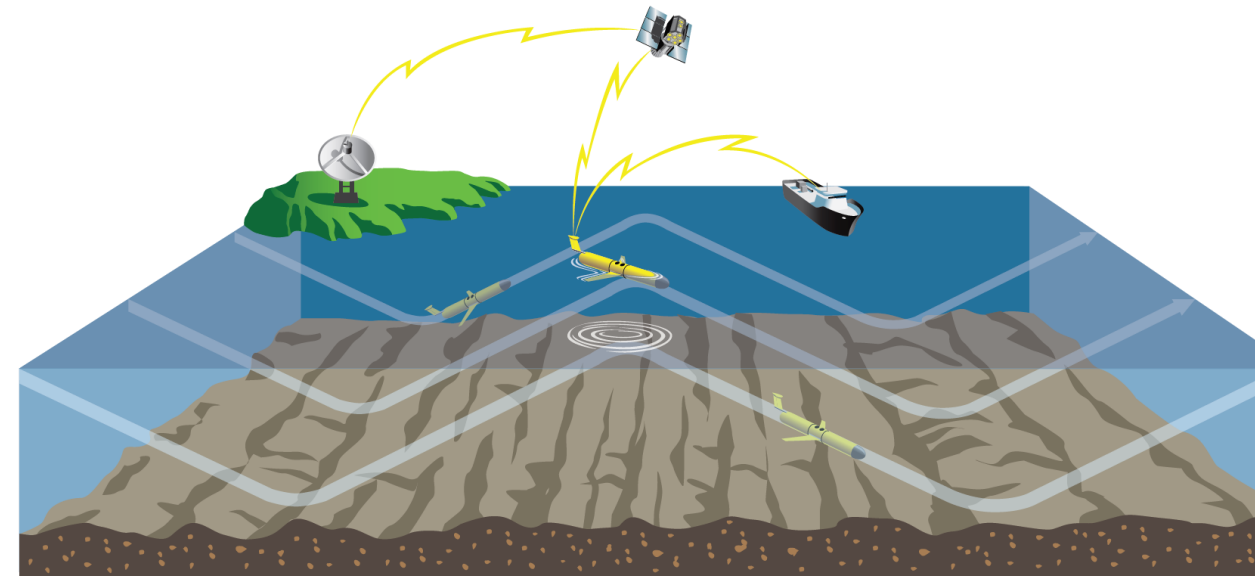
- Pitch battery position is measured in inches
- The pitch battery moves forward and aft to help the glider achieve the desired pitch angle
- Moving the pitch battery does not change the volume of vehicle!



# Glider Flight Fundamentals

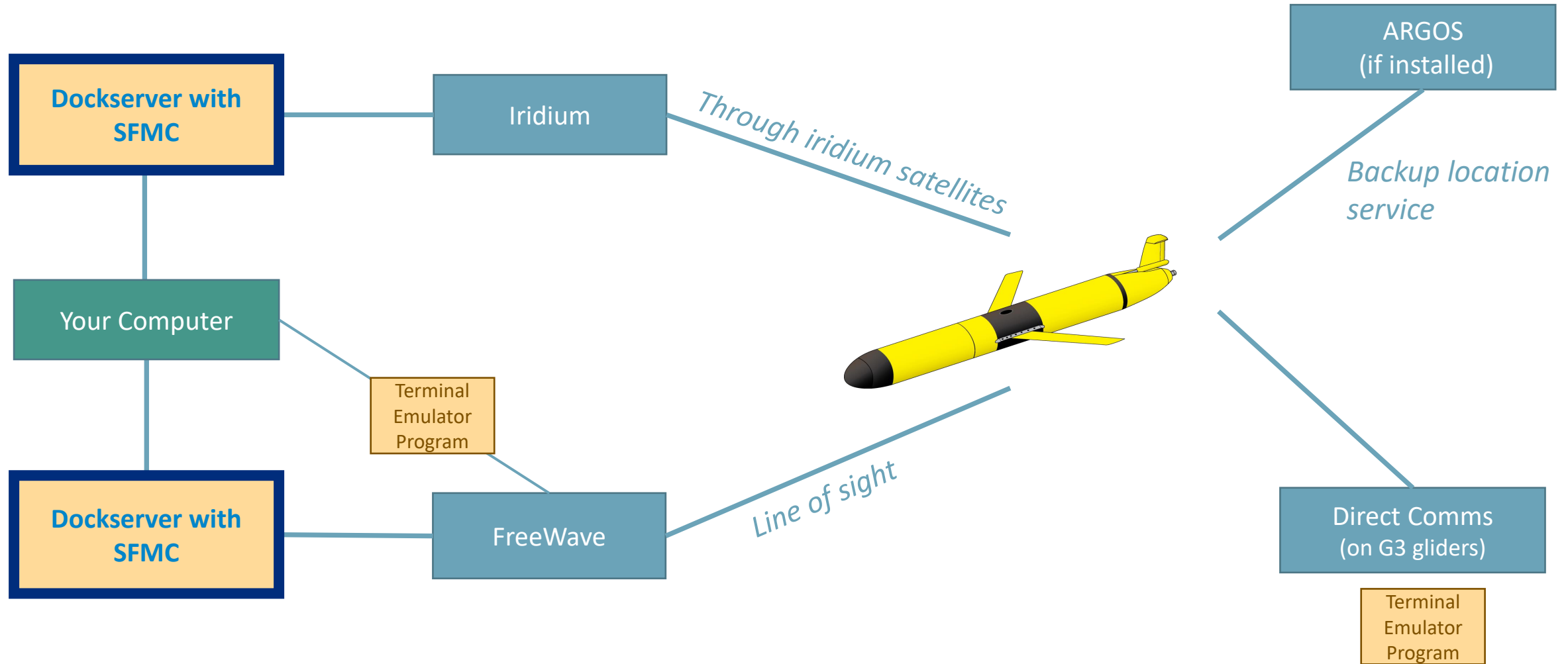
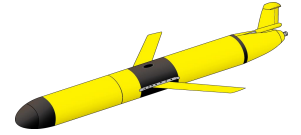


- Glider receives mission parameters
- Glider adjusts buoyancy and weight distribution to dive and surface, travelling in saw-tooth pattern
- Glider completes directed numbers of dives/surfaces (“yos”)
- Glider resurfaces, establishes GPS/comms, receives new mission parameters, transmits gathered data

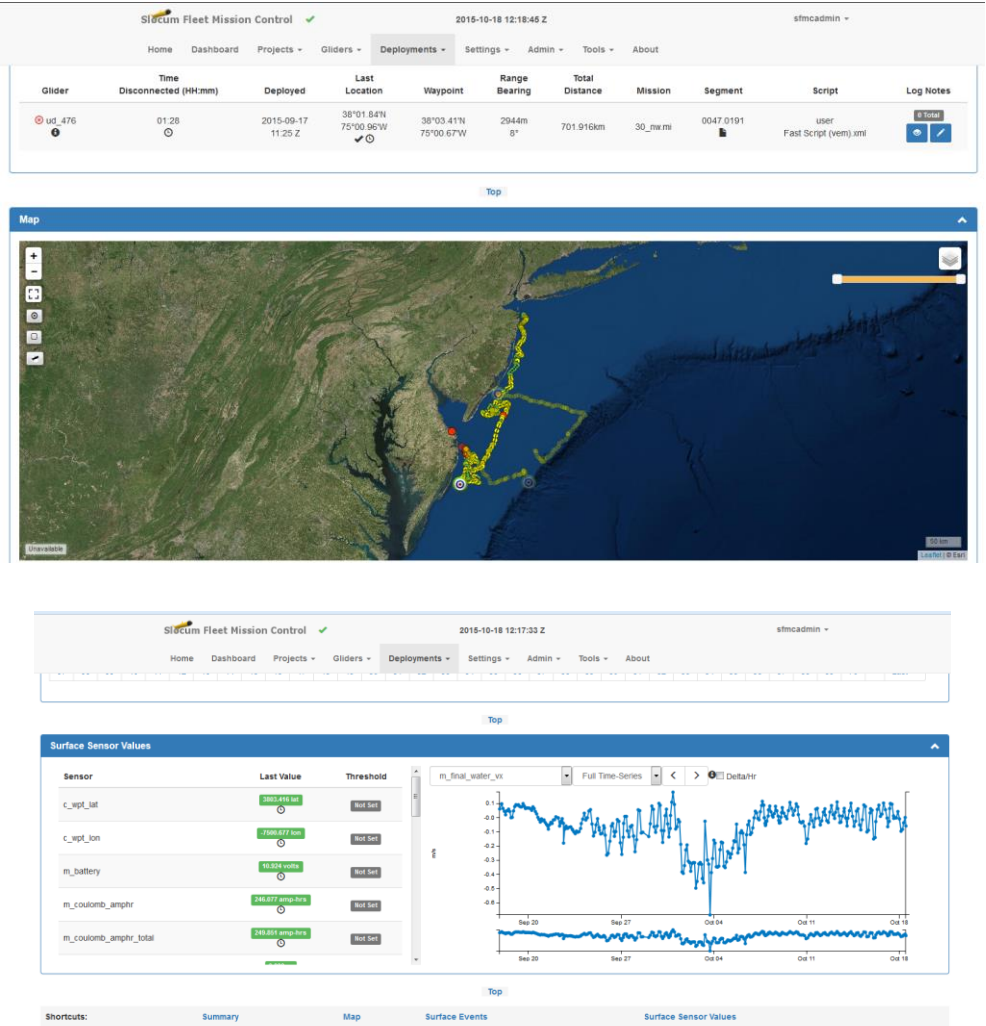
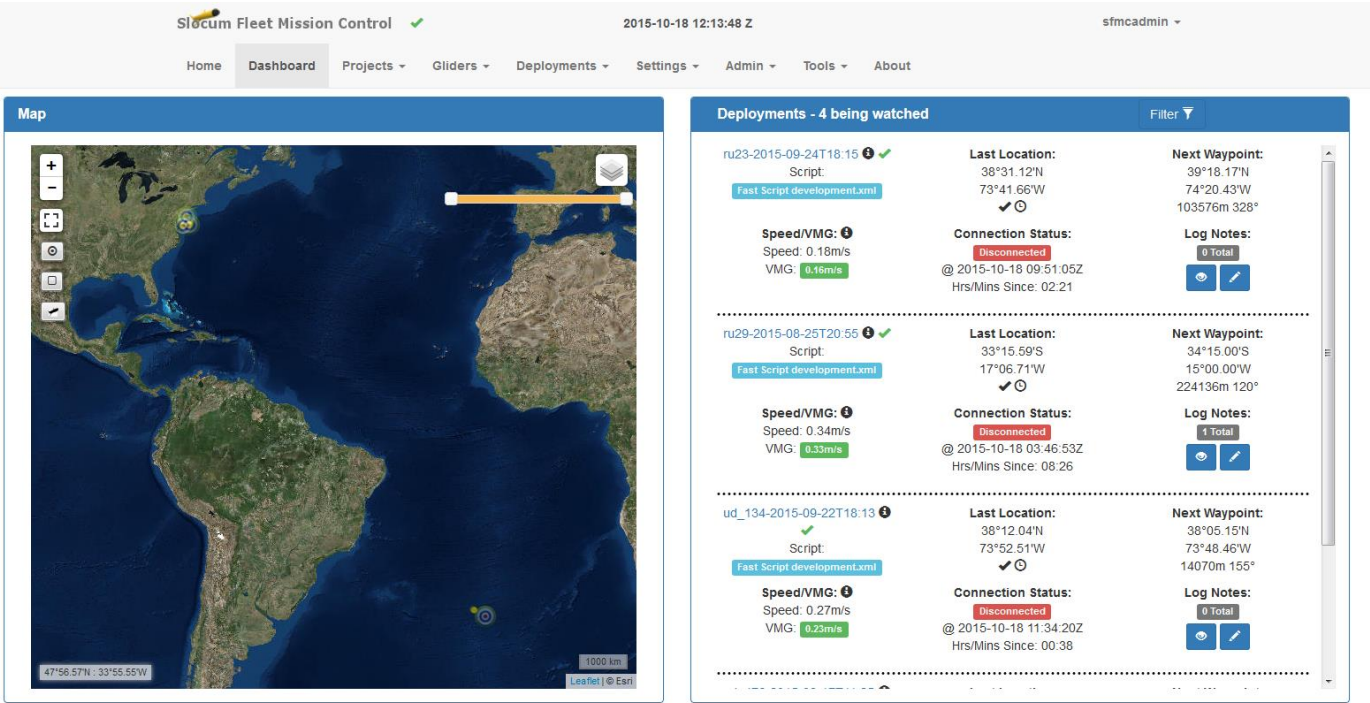


- Glider can communicate via Freewave comms any line-of-sight distance (up to 3 km)
- Glider can communicate via Iridium satellite to anywhere in the world

# Glider Communication



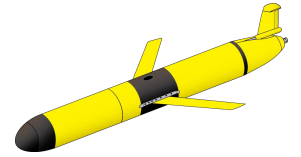
# Glider Software: Slocum Fleet Mission Control (SFMC)



Comprehensive Slocum Fleet Mission Control (SFMC) software suite allows glider tracking, mission planning, and piloting from around the world



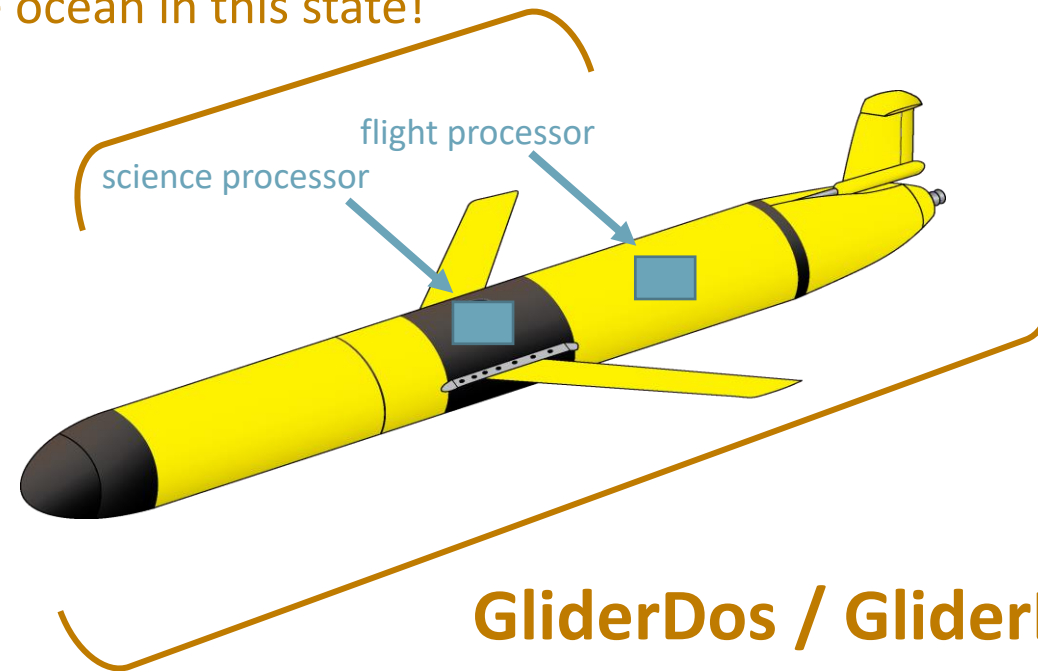
# Glider Operating States



**Pico** (persistor based glider) / **Shell** (stm32 based glider)

Only talking to the processors (base OS)

You do not want to be in the ocean in this state!

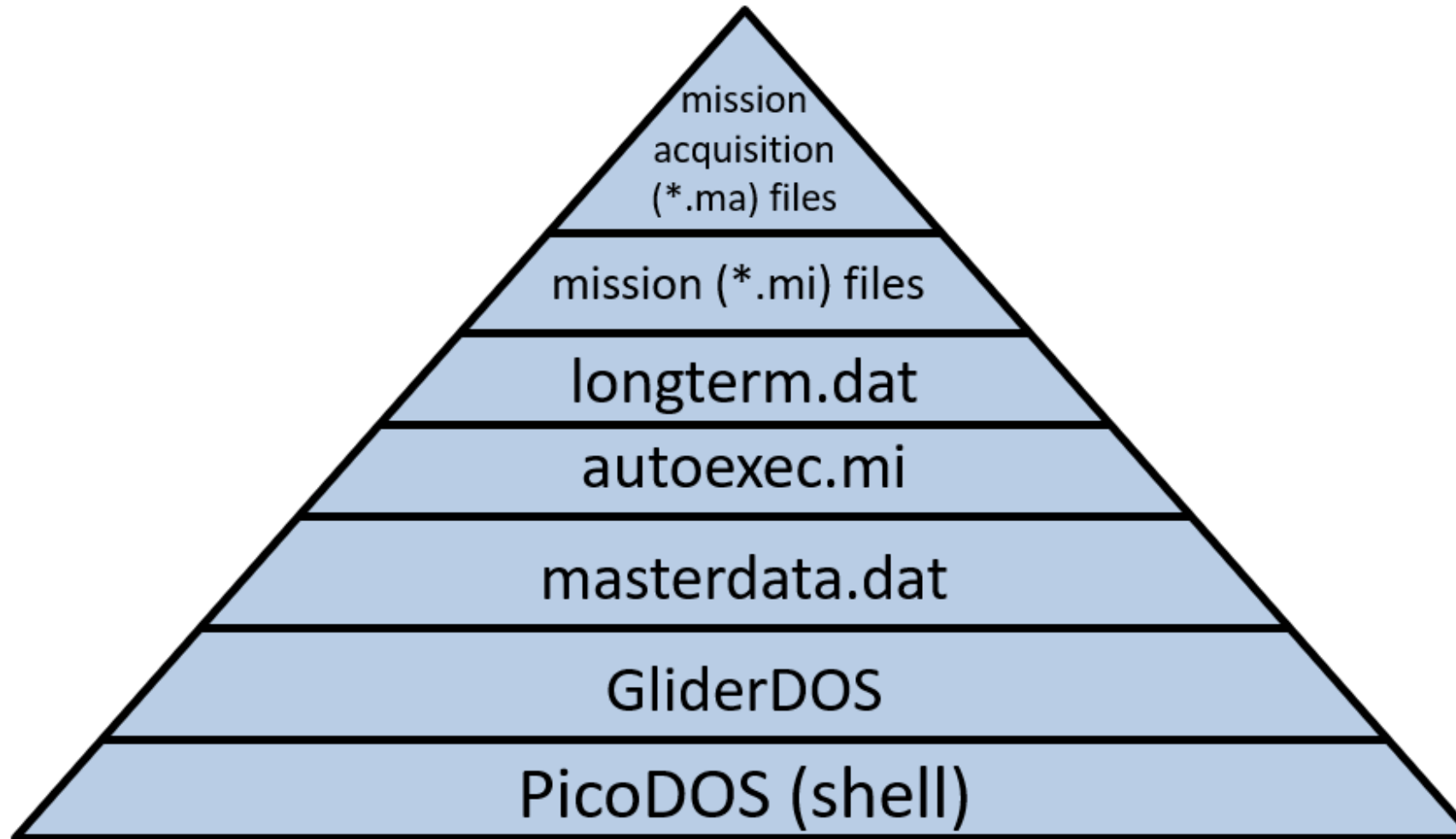
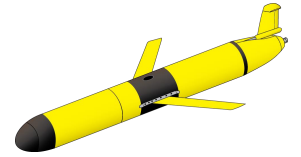


## GliderDos / GliderLab

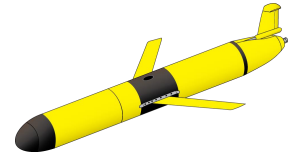
The processors are running the glider application

The glider can use important devices here

# Glider Software – *Hierarchy of Glider Software Needs*



# Glider Processors



- Flight Processor

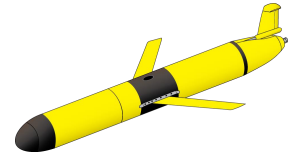
- glider.app (persistor) or flash-flight.gex (stm32)
- masterdata
- autoexec.mi
- longterm.dat
- missions
- ma files

-----Clothesline-----

- Science Processor

- supersci.app (persistor) or flash-science.gex (stm32)
- proglets.dat

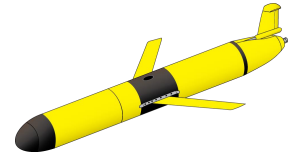
# Glider Variables — *Prefixes*



| Prefix | Significance                                   |
|--------|------------------------------------------------|
| m_     | Measured                                       |
| c_     | Commanded                                      |
| u_     | User-defined before run-time                   |
| f_     | Set in factory                                 |
| x_     | Computed at run time (never set these values!) |
| s_     | Simulated state variables                      |
| sci_   | Science sensor                                 |



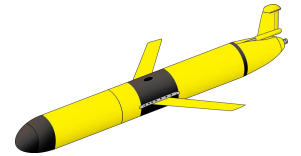
# Glider Data Files



|                                     | Flight Side | Science Side | New Compressed files |
|-------------------------------------|-------------|--------------|----------------------|
| ALL data                            | *.DBD       | *.EBD        | *.DCD / *.ECD        |
| Small data files<br>(user defined)  | *.SBD       | *.TBD        | *.SCD / *.TCD        |
| Medium data files<br>(user defined) | *.MBD       | *.NBD        | *.MCD / *.NCD        |
| Log files                           | *.MLG       | *.NLG        | *.MCG / *.NCG        |

- Logging on/off can create these 8 file types while in gliderdos or lab\_mode
- These log files are created at all times when mission is running (except when sending files or receiving files)
- SBD and TBD are customized to send over iridium comms and sent by SFMC script

# Glider Data Files — *Ascii Header*

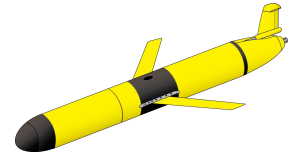


☰ capx594-2021-195-1-16.sbd ●

C: > Users > cgoodrich > Downloads > ☰ capx594-2021-195-1-16.sbd

```
1 dbd_label: DBD(dinkum_binary_data)file
2 encoding_ver: 5
3 num_ascii_tags: 14
4 all_sensors: F
5 the8x3_filename: 03100016
6 full_filename: capx594-2021-195-1-16
7 filename_extension: sbd
8 mission_name: sfmc.mi
9 fileopen_time: Thu_Jul_15_17:24:42_2021
10 total_num_sensors: 2678
11 sensors_per_cycle: 14
12 state_bytes_per_cycle: 4
13 sensor_list_crc: 88a952b1
14 sensor_list_factored: 1
15 sa4?y???B?i~T4o?Ad????(??*??(\?x?????"?y)?@鵒?wx??
16 ?<?A??@?+?=?? A??-m=d??P?@??w)?@Q?swx<
17 ?swx??/??F??
```

# Common Glider Commands

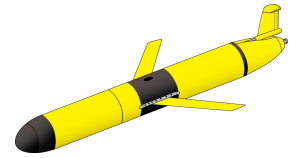


help  
lab\_mode  
wiggle  
use  
report ++  
get  
put  
! (bang)  
consci  
whoru  
where

zero\_ocean\_pressure  
dir  
run  
loadmission  
exit  
exit reset

**NOTE:** List all available commands  
by typing help

# Catching a Glider on Start Up



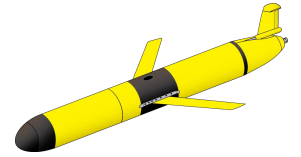
You should begin these commands when the glider says:

SEQUENCE: About to run initial.mi

- > **Ctrl-C** *# if successful you should see a GliderDos prompt*
- > **callback 30** *# this will hang up the iridium phone for 30 mins.*
- > **lab\_mode on** *# only if you are in the lab!*
- > **ballast** *# only if you are in the lab!*
- > **report clearall** *# to clear buoyancy pump output from 'ballast'*



# File Transfer Commands – *from the glider*



## *zs*

> `zs path/fileName`

- Glider sends file from specified path (from the processor that you are currently talking to)

## *szs*

> `szs path/fileName`

- Glider sends file from specified path on the science processor (can only be issued from flight processor)

## *send*

> `send logFileNames`

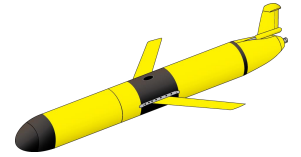
- Glider sends specified log files – path not specified – files are moved to sentlogs once the transfer is complete

## *s*

> `s logFileNames`

- Same functionality as ‘send’ but to be used when a glider is in a mission

# File Transfer Commands – *to the glider*



***zr***

> zr

- Glider receives file - only to be used when not using a dockserver – user selects file that the glider will receive (file lands on processor you are talking to)

***dockzr***

> dockzr fileName

- Glider receives file from 'to-glider' folder on the dockserver - only to be used with a dockserver (file lands on processor you are talking to)

***szr***

> szr

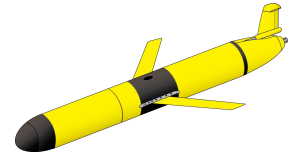
- Science processor receives file - only to be used when not using a dockserver – user selects file that the glider will receive (can only be issued from flight processor)

***dockszr***

> dockszr fileName

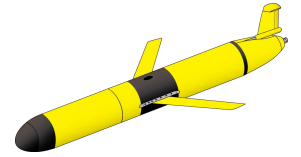
- Science processor receives file from 'to-science' folder on the dockserver - only to be used with a dockserver (can only be issued from flight processor)

# Config Directory – *glider configuration files*



- config.srf
  - Customize your surface dialog
- sbdlist.dat
  - Customize content of SBD file
- tbdlist.dat
  - Customize content of TBD file (sits on science processor)
- longterm.dat
  - Customize list of sensors that are stored whenever glider powers down

# Surface Dialog – *output when the glider ‘surfaces’*



Defined in the file config.srf – can be customized

Slucum Fleet Mission Control 2021-09-02 21:04:54 cgoodrich

Mission Control Configuration History Tools Admin About

Glider Terminal Access / Terminal for Glider: capx594 / Options

Other Connected Gliders

Other Disconnected Gliders

>\_blue

>\_capex690

>\_capex690-1

>\_capx594-1

>\_capx638

>\_g3s\_capex753

>\_g3s\_capex753\_hd

>\_glider01

>\_gn\_537

00:00 Terminal for Glider: capx594

modem:/dev/tty\_dgrp\_gl\_0

Glider capx594 at surface.  
Because:specified UTC time [behavior surface\_7 start\_when = 13.0]  
MissionName:astock.mi MissionNum:capx594-2021-232-0-186 (0324.0186)  
Vehicle Name: capx594  
Curr Time: Thu Sep 2 15:09:57 2021 MT: 1078235  
DR Location: 4159.338 N -7018.399 E measured 121.043 secs ago  
GPS TooFar: 69696969.000 N 69696969.000 E measured 1e+308 secs ago  
GPS Invalid : 4159.185 N -7018.121 E measured 207.407 secs ago  
GPS Location: 4159.338 N -7018.399 E measured 122.102 secs ago  
sensor:c\_autoballast\_state(enum)=2 814.283 secs ago  
sensor:c\_climb\_bpump(X)=135 814.283 secs ago  
sensor:c\_dive\_bpump(X)=-225 814.287 secs ago  
sensor:c\_iridium\_current\_num(enum)=1 14061 secs ago  
sensor:c\_wpt\_lat(lat)=4200.6 2065.14 secs ago  
sensor:c\_wpt\_lon(lon)=-7024.52 2065.15 secs ago  
sensor:m\_battery(volts)=14.2714484171615 47.242 secs ago  
sensor:m\_bms\_aft\_current(amp)=0.258752 3.412 secs ago  
sensor:m\_bms\_ebay\_current(amp)=-0.00125 3.443 secs ago  
sensor:m\_bms\_pitch\_current(amp)=0.203752 3.475 secs ago  
sensor:m\_coulomb\_amphr(amp-hrs)=140.157277999998 3.332 secs ago  
sensor:m\_coulomb\_amphr\_total(amp-hrs)=140.387279999996 3.336 secs ago  
sensor:m\_digifin\_leakdetect\_reading(nodim)=1021 7.574 secs ago  
sensor:m\_iridium\_attempt\_num(nodim)=2 56.55 secs ago  
sensor:m\_iridium\_signal\_strength(nodim)=5 43.58 secs ago  
sensor:m\_leakdetect\_voltage(volts)=2.49435286935287 39.228 secs ago  
sensor:m\_leakdetect\_voltage\_forward(volts)=2.48797313797314 39.193 secs ago  
sensor:m\_leakdetect\_voltage\_science(volts)=2.48730158730159 39.158 secs ago  
sensor:m\_lithium\_battery\_relative\_charge(%)=34.7035906976763 3.366 secs ago

Files

from-glider (1576)

to-glider (0)

To Glider File Drop Zone

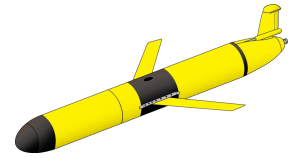
to-science (0)

To Science File Drop Zone

logs (302)

archive (18)

# sbdlist.dat – defines data in \*.sbd files



## Customize content (variables and decimation) :

|                             |      |    |    |    |
|-----------------------------|------|----|----|----|
| m_present_secs_into_mission |      |    |    |    |
| m_present_time              |      |    |    |    |
| m_water_vx                  | 0    | 15 | -1 | -1 |
| m_water_vy                  |      |    |    |    |
| m_de_oil_vol                | 60   | 15 | -1 | -1 |
| c_de_oil_vol                | 60   |    |    |    |
| m_ballast_pumped            | 120  | 15 | -1 | 2  |
| c_ballast_pumped            | 120  | 15 | -1 | 2  |
| m_gps_lat                   | 300  |    |    |    |
| m_gps_lon                   |      |    |    |    |
| m_heading                   | 300  |    |    |    |
| m_lat                       | 600  |    |    |    |
| m_lon                       | 600  |    |    |    |
| m_pitch                     | 60   | 15 | -1 | 2  |
| m_roll                      | 60   | 15 | -1 | 2  |
| c_wpt_lat                   |      |    |    |    |
| c_wpt_lon                   |      |    |    |    |
| m_battery                   | 3600 |    |    |    |
| m_battpos                   | 30   |    |    |    |
| m_depth                     | 40   |    |    |    |

Must have time variables

Default decimation

If decimation not specified, default will be used

Variable

Interval (sec)

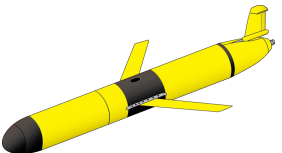
Yo\_duty\_cycle

State

Halfyos

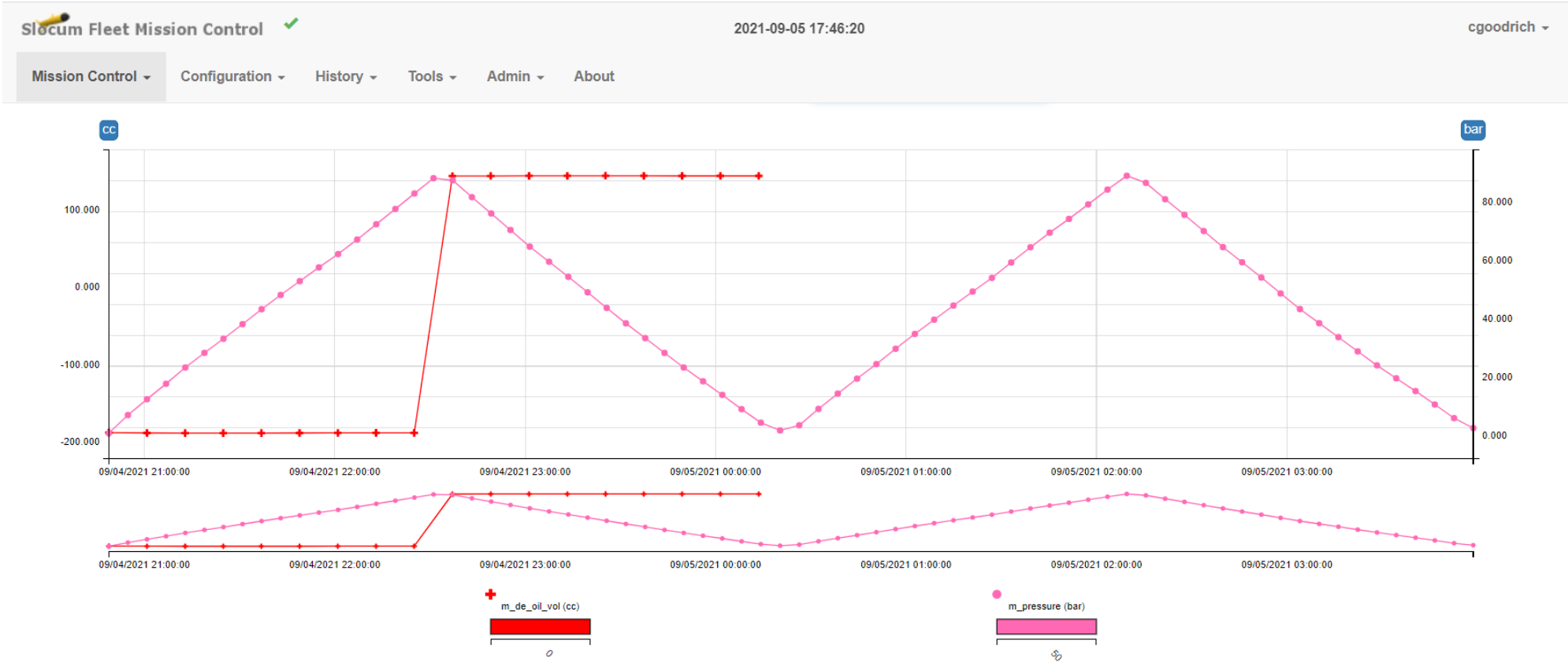


# sbdlist.dat – defines data in \*.sbd files

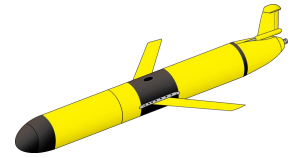


**Example:**

|              |     |   |   |
|--------------|-----|---|---|
| m_pressure   | 360 | 7 |   |
| m_de_oil_vol | 720 | 7 | 2 |



# tbdlist.dat – defines data in \*.tbd files



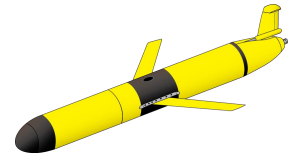
## Customize content (variables and decimation) :

|                                 |    |    |   |
|---------------------------------|----|----|---|
| SCI_M_PRESENT_TIME              |    |    |   |
| SCI_M_PRESENT_SECS_INTO_MISSION |    |    |   |
| SCI_WATER_COND                  | 30 | 15 | 3 |
| SCI_WATER_TEMP                  | 30 | 15 | 3 |
| SCI_WATER_PRESSURE              | 30 | 15 | 3 |

Must have time variables

- Same syntax as sbdlist.dat but only for science variables
- Sits on science processor (c:/config/tbdlist.dat)

# longterm.dat – *variables saved when glider powers down*



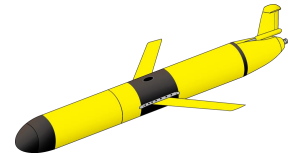
Oil bladders/bellaframs  
require replacement  
every 20,000 half cycles

Glider's "fuel gauge"

```
m_avg_climb_rate
m_avg_upward_inflection_time
★ m_tot_num_inflections
m_tot_horz_dist
m_lat
m_lon
m_tot_ballast_pumped_energy
m_battery
m_iridium_call_num
m_iridium_dialed_num
★ m_coulomb_amphr_total
s_water_depth_avg
s_water_depth_delta
s_water_depth_wavelength
f_ocean_pressure_min
m_avg_speed
x_last_wpt_lat
x_last_wpt_lon
x_de_avg_oil_vol_ierr_on_ascent
x_de_avg_oil_vol_ierr_on_descent
x_hover_ballast_shallow
x_hover_ballast_deep
x_hover_depth_shallow
x_hover_depth_deep
```

- ***Customize list of sensors that are stored whenever glider powers down***

# Installed Devices



- Devices are defined in at the beginning of autoexec.mi
- 'use' command or 'Ctrl-W' in a mission

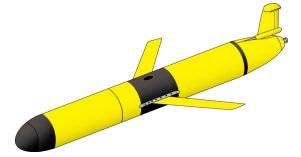
```

name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
[I Installed] [- Not_Installed]
[u In_use] [- Not_In_use] [X out_of_Service]
limits stats (#total/#mission/#segment)

0   simdrv -
1   test_driver -
2   ARGOS* I u -1 20 5 0
3   WATCHDOG I u -1 -1 -1 0
4   DEADMAN I u -1 20 5 0
5   CONSOLE* I u -1 20 5 0
6   GPS I u -1 20 5 0
7   pinger -
8   attitude -
9   attitude_tcm3 -
10  attitude_rev I u 3 20 5 0 [ 0 0 0] [ 1 0 0] [ 0 0 0]
11  ocean_pressure I u 3 20 5 0
12  vacuum I u 3 20 5 0
13  battery I u 3 20 5 0
14  lithium_battery -
15  air_pump I u 3 20 5 0
16  pitch_motor I u 3 20 5 0
17  science_super I u 3 20 5 0
18  roll_motor -
19  fpitch_pump -
20  fin_motor -
21  digifin I u 3 20 5 0
22  altimeter I u 3 20 5 0
23  ctd -
24  IRIDIUM* I u -1 20 5 0 [ 0 0 0] [ 0 0 0] [ 2 0 0]
25  leakdetect I u 3 20 5 0
26  recovery I u 3 20 5 0
27  coulomb I u 3 20 5 0
28  veh_temp I u 3 20 5 0
29  BUOYANCY_PUMP I u 3 20 3 0
30  THERMAL_ACC_PRE -
31  THERMAL_ENGINE -
32  THERMAL_PUMP -
33  DE_PUMP -
34  thruster_g1 -
35  thruster I u 3 20 5 0
devices:(t/m/s) errs: 0/ 0/ 0 warn: 1/ 0/ 0 odd: 2/ 0/ 0

```

# Installed Devices



```

name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
[I Installed] [- Not_Installed]
[u In_use] [- Not_In_use] [X Out_of_Service]
name          limits      stats (#total/#mission/#segment)
              os w/s w/m os      errors      warnings      oddities
0      simdrv  I u    3  20   5  0
1    test_driver I u    3  20   5  0
2      ARGOS*  I u   -1  20   5  0
3    WATCHDOG  I u   -1 -1  -1  0
4    DEADMAN   I u   -1  20   5  0
5    CONSOLE*  I u   -1  20   5  0
6      GPS     I u   -1  20   5  0
7  attitude_rev I u    3  20   5  0
8 ocean_pressure I u    3  20   5  0
9      vacuum  I u    3  20   5  0
10     battery I u    3  20   5  0
11    air_pump I u    3  20   5  0
12  pitch_motor I u    3  20   5  0
13 science_super I u    3  20   5  0
14    digifin  I u    3  20   5  0
15    altimeter I u    3  20   5  0
16 altimeter_232 -
17    IRIDIUM* I u   -1  20   5  0 [  0  0  0] [  0  0  0] [  2  2  2]
18    leakdetect I u    3  20   5  0
19    recovery  -
20    coulomb   I u    3  20   5  0
21    veh_temp  I u    3  20   5  0
22 BUOYANCY_PUMP I u    3  20   3  0
23    DE_PUMP  -
24    HD_PUMP  -
25    thruster  I u    3  20   5  0
devices:(t/m/s) errs:  0/  0/  0 warn:  0/  0/  0 odd:  2/  2/  2

```

Type `use - sensor_name` to temporarily remove devices

Type `use + sensor_name` to reinstall devices that have been taken out of service

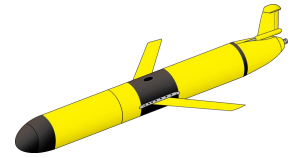
Type `use all` or `use none` to install or remove all devices





← 'u' indicates in use

# Installed Devices



```

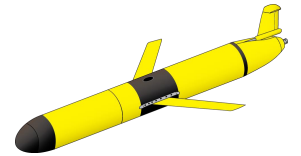
name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
[I Installed] [- Not_Installed]
[u In_use] [- Not_In_use] [X out_of_Service]
limits stats (#total/#mission/#segment)

0  simdrv -
1  test_driver -
2  ARGOS* I u -1 20 5 0
3  WATCHDOG I u -1 -1 -1 0
4  DEADMAN I u -1 20 5 0
5  CONSOLE* I u -1 20 5 0
6  GPS I u -1 20 5 0
7  pinger -
8  attitude -
9  attitude_tcm3 -
10 attitude_rev I u 3 20 5 0 [ 0 0 0] [ 1 0 0] [ 0 0 0]
11 ocean_pressure I u 3 20 5 0
12 vacuum I u 3 20 5 0
13 battery I u 3 20 5 0
14 lithium_battery -
15 air_pump I u 3 20 5 0
16 pitch_motor I u 3 20 5 0
17 science_super I u 3 20 5 0
18 roll_motor -
19 fpitch_pump -
20 fin_motor -
21 digifin I u 3 20 5 0
22 altimeter I u 3 20 5 0
23 ctd -
24 IRIDIUM* I u -1 20 5 0 [ 0 0 0] [ 0 0 0] [ 2 0 0]
25 leakdetect I u 3 20 5 0
26 recovery I u 3 20 5 0
27 coulomb I u 3 20 5 0
28 veh_temp I u 3 20 5 0
29 BUOYANCY_PUMP I u 3 20 3 0
30 THERMAL_ACC_PRE -
31 THERMAL_ENGINE -
32 THERMAL_PUMP -
33 DE_PUMP -
34 thruster_g1 -
35 thruster I u 3 20 5 0
devices:(t/m/s) errs: 0/ 0/ 0 warn: 1/ 0/ 0 odd: 2/ 0/ 0
    
```

Annotations for the IRIDIUM\* row (line 24):

- Total # of warnings:** Points to the first bracketed array `[ 0 0 0]`.
- # of warnings in segment:** Points to the second bracketed array `[ 0 0 0]`.
- # of warnings in mission:** Points to the third bracketed array `[ 2 0 0]`.

# Installed Devices



```

name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
[I Installed] [- Not_Installed]
[u In_use] [- Not_In_use] [X out_of_Service]
limits stats (#total/#mission/#segment)

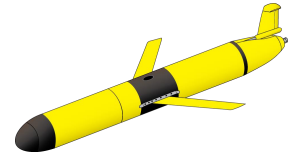
0  simdrv -
1  test_driver -
2  ARGOS* I u -1 20 5 0
3  WATCHDOG I u -1 -1 -1 0
4  DEADMAN I u -1 20 5 0
5  CONSOLE* I u -1 20 5 0
6  GPS I u -1 20 5 0
7  pinger -
8  attitude -
9  attitude_tcm3 -
10 attitude_rev I u 3 20 5 0 [ 0 0 0] [ 1 0 0] [ 0 0 0]
11 ocean_pressure I u 3 20 5 0
12 vacuum I u 3 20 5 0
13 battery I u 3 20 5 0
14 lithium_battery -
15 air_pump I u 3 20 5 0
16 pitch_motor I u 3 20 5 0
17 science_super I u 3 20 5 0
18 roll_motor -
19 fpitch_pump -
20 fin_motor -
21 digifin I u 3 20 5 0
22 altimeter I u 3 20 5 0
23 ctd -
24 IRIDIUM* I u -1 20 5 0 [ 0 0 0] [ 0 0 0] [ 2 0 0]
25 leakdetect I u 3 20 5 0
26 recovery I u 3 20 5 0
27 coulomb I u 3 20 5 0
28 veh_temp I u 3 20 5 0
29 BUOYANCY_PUMP I u 3 20 3 0
30 THERMAL_ACC_PRE -
31 THERMAL_ENGINE -
32 THERMAL_PUMP -
33 DE_PUMP -
34 thruster_g1 -
35 thruster I u 3 20 5 0
devices:(t/m/s) errs: 0/ 0/ 0 warn: 1/ 0/ 0 odd: 2/ 0/ 0
    
```

Total # of oddities

# of oddities in segment

# of oddities in mission

# Mission Planning



## Ballast & H-Moment

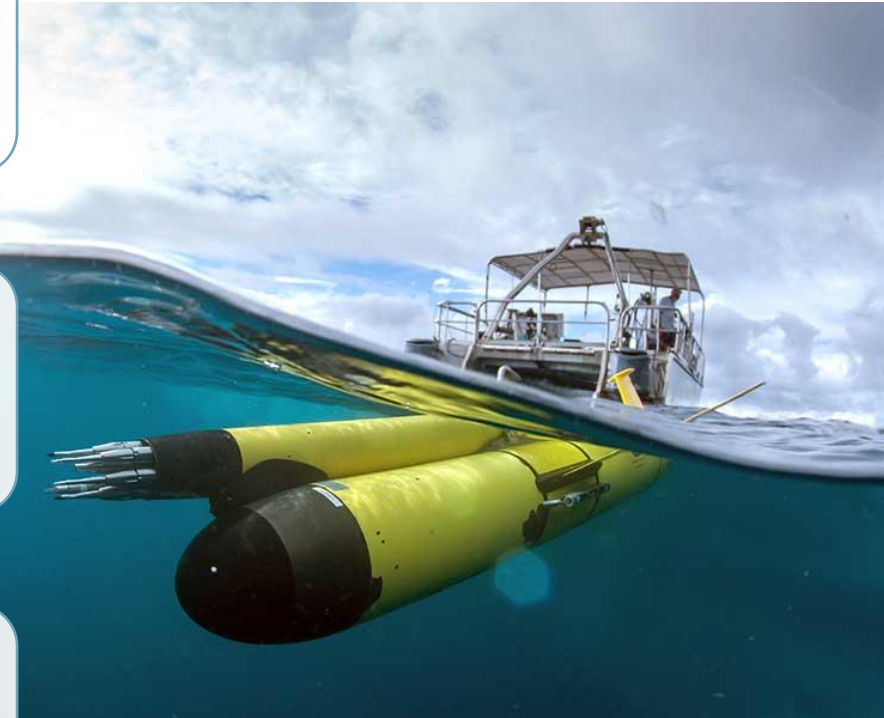
- Identify surface Temp & Salinity conditions at deployment site
- $5 < \text{H-moment} < 6$  (ideal)
- Final Seal

## Testing & Functional

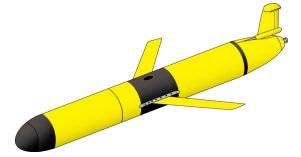
- Vacuum test
- Functional checkout procedure

## Simulate Missions

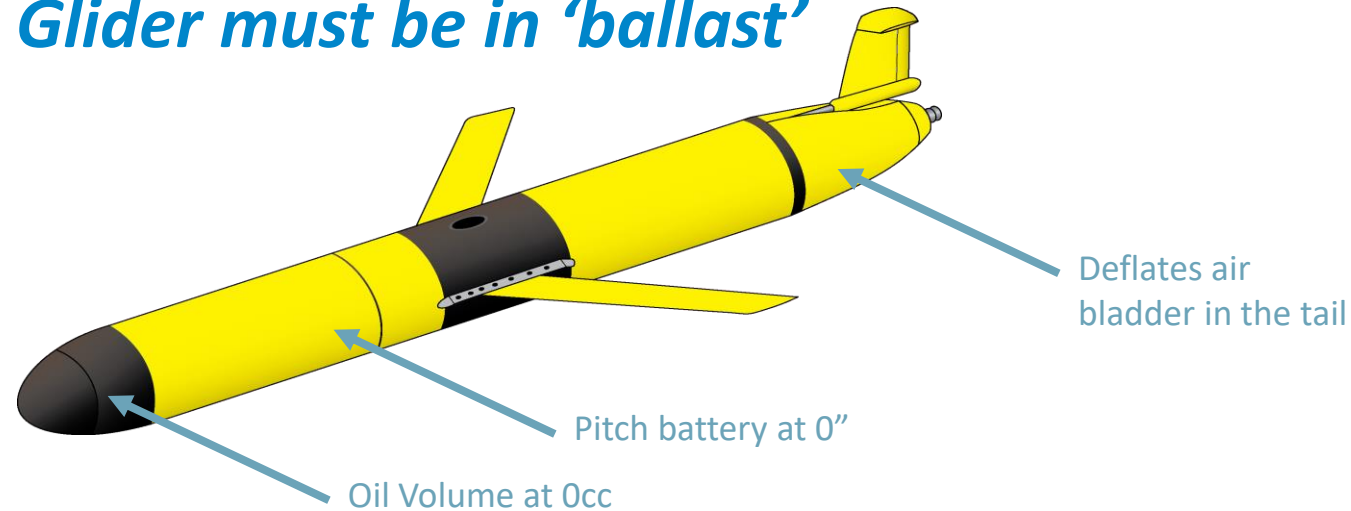
- Easy: modify existing missions
- Harder & riskier without experience: write your own



# Ballasting



*Glider must be in 'ballast'*



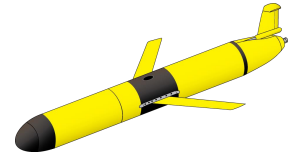
Goal is to have the glider neutrally buoyant for the surface of the target water.

Need to know Temp & Salinity/Cond/Density of where the glider is being deployed

$\rho$  = density  
 $m$  = mass  
 $V$  = volume

$$\rho = \frac{m}{V}$$

# Mission Commands – *how to start a glider mission*



## *loadmission*

- > `loadmission mission1.mi`
  - Sets glider sensor values found in mission1.mi (does not start an actual mission)

## *run*

- > `run mission2.mi`
  - Instructs glider to begin running mission2.mi

## *sequence*

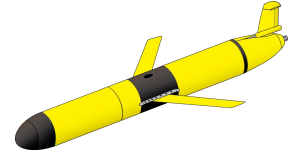
- > `sequence mission3.mi mission4.mi mission5.mi`
  - Instructs glider to run mission3.mi. When mission3.mi completes glider will run mission4.mi, and when that finishes it will run mission5.mi

**OR**

- > `Sequence mission5.mi (5)`
  - Instructs glider to run mission5.mi five times in a row



# Testing Your New Glider



- Make sure to confirm the following:

- FreeWave comms
- Iridium comms
- Wiggle (exercise all motors)
- Science sensor output

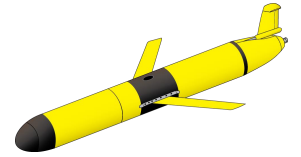
Functional Checkout Procedure  
(4095-FCP)

- Personalize your glider!

- Rename it
- Add your phone numbers
- Connect to your dockserver



# Glider Dos & Don'ts



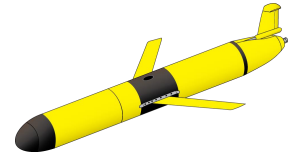
## DO

- Secure it properly in crate with all three straps for shipping.
- Use fresh desiccants on each deployment.
- Monitor internal vacuum before launch (less vacuum indicates a leak; positive pressure may indicate dangerous gas accumulation).
- Simulate missions before launch.
- Test Iridium and Argos telemetry before launch.
- Burn off passivation in lithium batteries with a ~500m dive

## DON'T

- Never power up a shallow glider without a vacuum.
- Never run a simulation on a glider other than “on\_bench.”
- Never deploy a glider in simulation.
- Never deploy a glider in “boot pico”(persistor) or “boot shell”(stm32)
- Never exit to pico (persistor) or exit to shell (stm32) during a deployment.
- Never power on a glider with more than 15 vDC from an external power supply.
- Never deploy a glider in lab\_mode.
- Never perform the top of a yo below 30 meters (with 100 or 200 meter glider).
- Never secure the glider to the cart while over railing or in the water.

# Water Space Management

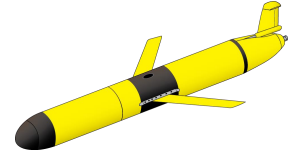


Consider the following when planning your deployment:

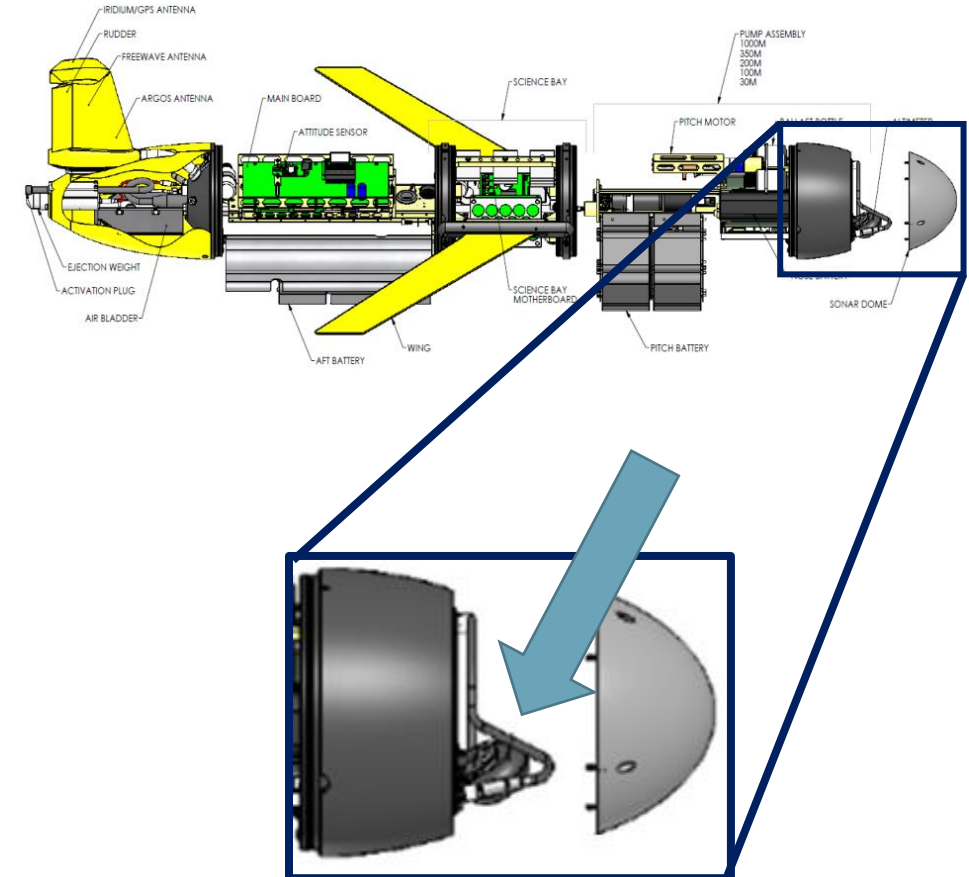
- Tides
- Shipping lanes
- Recreational traffic
- Fishing grounds
- Bathymetry & sea floor characteristics



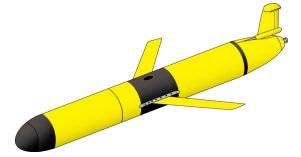
# Altimeters



- The altimeter is downward looking
- Avoid flying close to bottom in dynamic terrain
- Altimeters use energy – Pilots might choose to turn off when not needed:
- sensor: `u_alt_min_depth(m)` 2.0  
#how deep vehicle must be to use altitude



# Battery Options



## Alkaline Batteries

- Most Affordable Option
- Least Endurance
- Lightest (easy to ballast)
- Number of batteries may vary depending on payload

## Lithium Primary Batteries

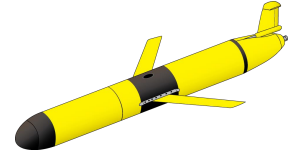
- Highest Endurance
- 3s (12V) and 4s (15V) options
- Requires Enable Circuit
- Proper shipping certification needed for Hazardous goods

## Lithium Rechargeable Batteries

- Rechargeable Battery Option
- Requires Battery Management System Board
- Proper shipping certification needed for Hazardous goods

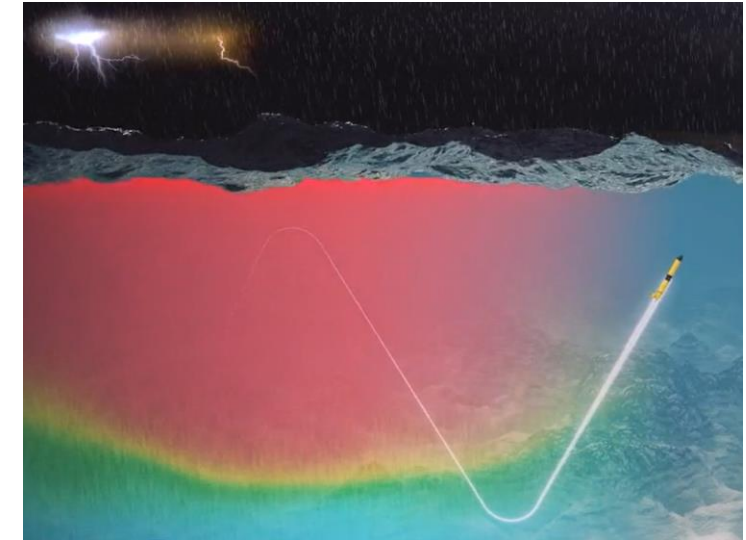
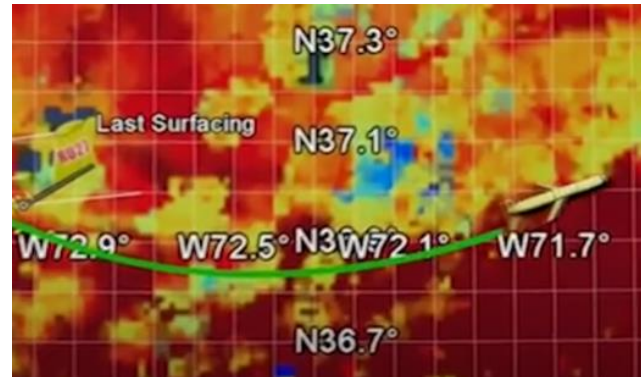
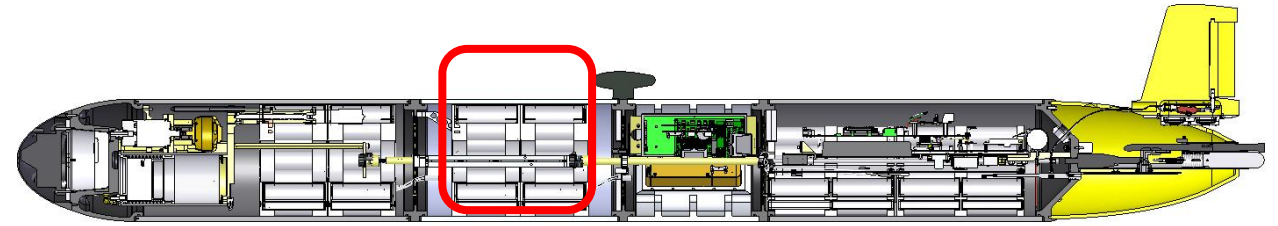


# Glider Endurance



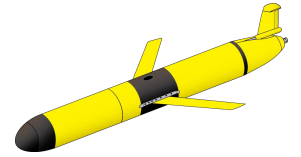
What factors come into play?

- Glider Build
  - Which batteries used?
  - Extended endurance payload?
  - Sensors installed
- Mission length
  - Power consumption
- Sampling strategy
  - How frequently do you need science data?
- Do you need to use the altimeter?
  - Can be uninstalled in deep water
  - Remember not to overwrite altimeter settings in mission





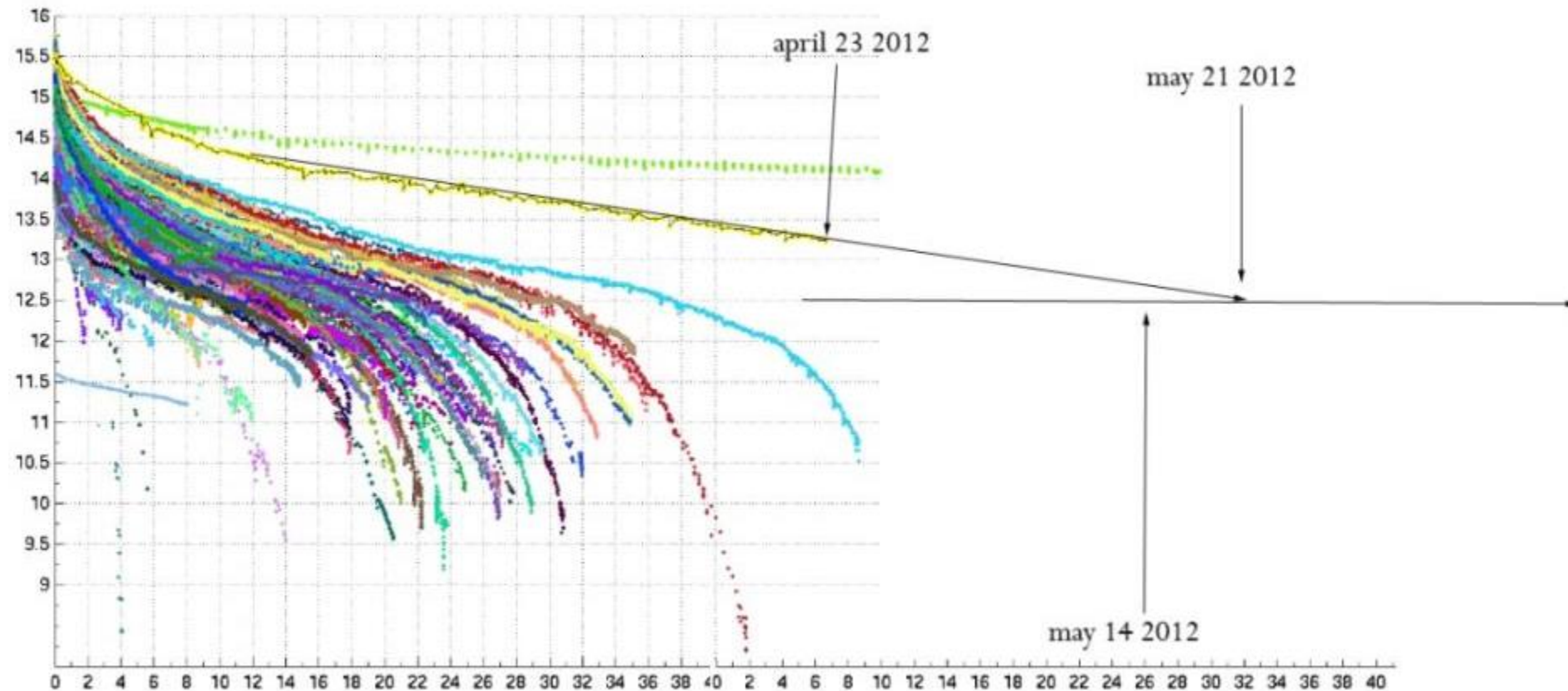
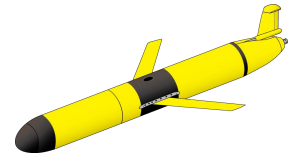
# Coulomb Counter and Battery Capacities



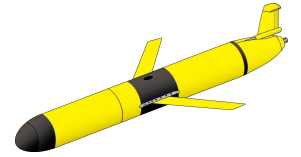
- The coulomb counter is the glider's "fuel tank"
- Important for lithium battery sets.
  - Lithium Primary 3s batteries have a nominal value of 720 amp-hrs
  - Lithium Primary 4s batteries have a nominal value of 550 amp-hrs
  - Gliders will begin aborting for low remaining energy when `m_coulomb_amphr_total` reaches 10% remaining)
  - zero this value when batteries are replaced:  
`put m_coulomb_amphr_total 0`
- User must set `f_coulomb_battery_capacity` for each battery type (typically done in `autoexec.mi`)

|                               | Total Non-derated<br>Amphrs | Total Non-derated<br>Watt Hours | Recommended<br>f_coulomb_battery_<br>capacity(amp-hrs) | Recommended<br>Undervolts(V) |
|-------------------------------|-----------------------------|---------------------------------|--------------------------------------------------------|------------------------------|
| Alkaline Standard             | 168                         | 2520                            | 120                                                    | 10                           |
| 3s Primary Standard           | 780                         | 8424                            | 720                                                    | 9.8                          |
| 3s Primary Extended           | 1140                        | 12312                           | 1050                                                   | 9.8                          |
| 4s Primary Standard           | 600                         | 8600                            | 550                                                    | 12                           |
| 4s Primary Extended           | 870                         | 12500                           | 800                                                    | 12                           |
| Lithium Rechargeable Standard | 234.6                       | 3300                            | 215                                                    | 12.5                         |
| Lithium Rechargeable Extended | 326.4                       | 4700                            | 300                                                    | 12.5                         |

# Voltage Curve – *alkaline*

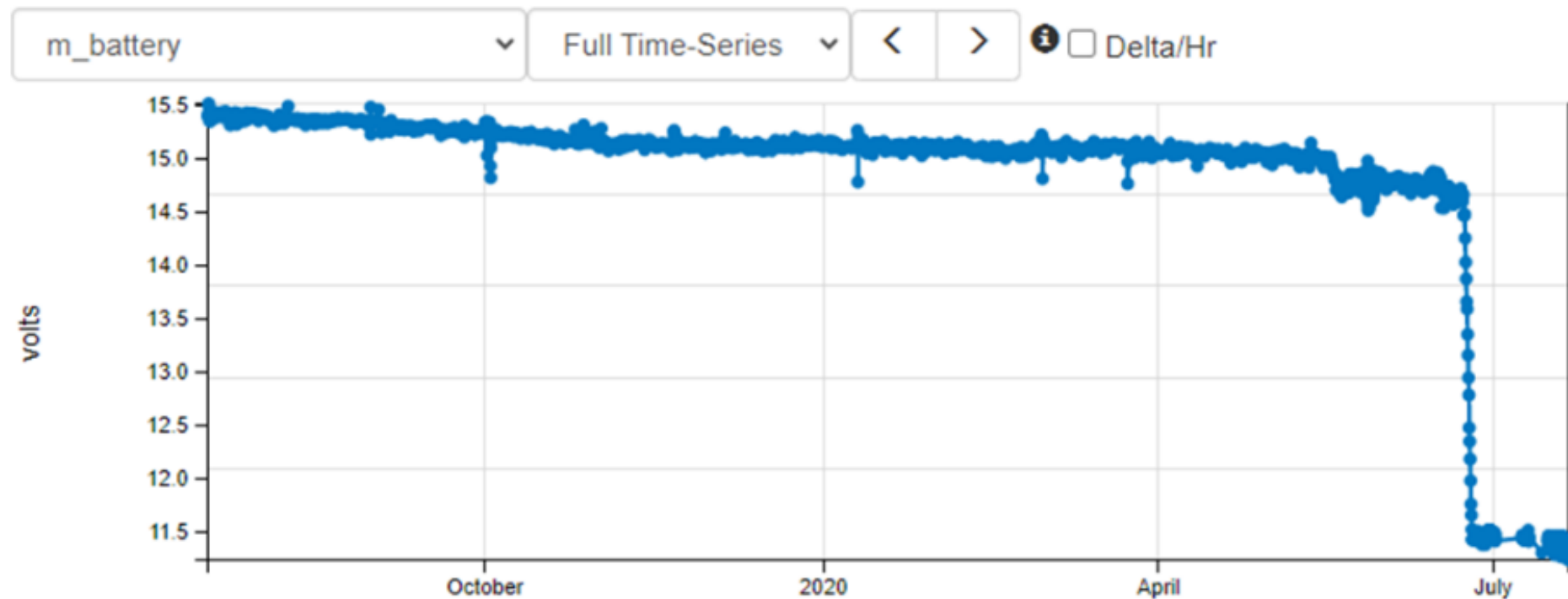


# Voltage Curve – 4S lithium batteries

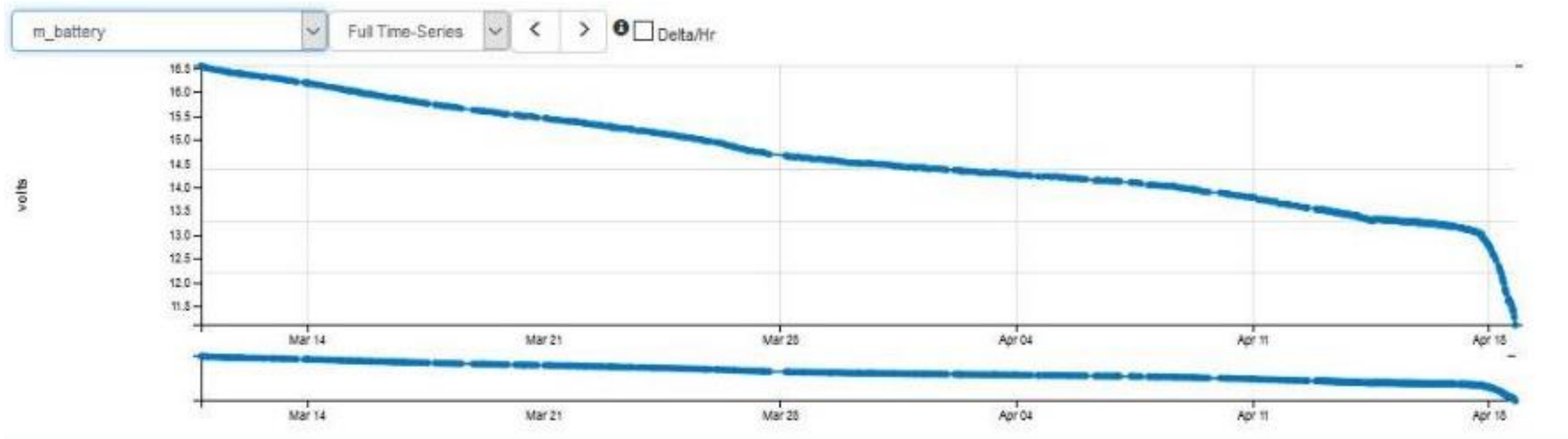
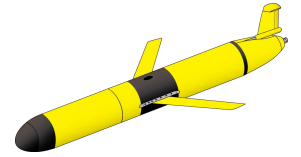


## Example of 4S voltage curve

(this glider was on a low power mission with energy bay)

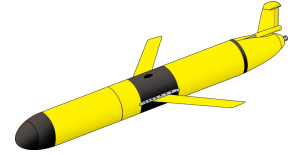


# Voltage Curve – *Rechargeable lithium batteries*



*221 total coulomb amphr when they shut off a 10v*

# Crisis and Mission End Management



## Safely getting your Glider home

Make sure you have answers to the following questions when it comes time to recover your glider:

- Who?
  - Which members of your team are responsible and available to recover the glider?
  - Who is backup? Who can help?
- Where?
  - Is your glider in an easy-to-recover area?
  - How will you get to/from this area?
- When?
  - When will the recovery occur?
  - What is your plan if the glider aborts early?
- How?
  - What resources / ships / boats etc. are needed to recover the glider?

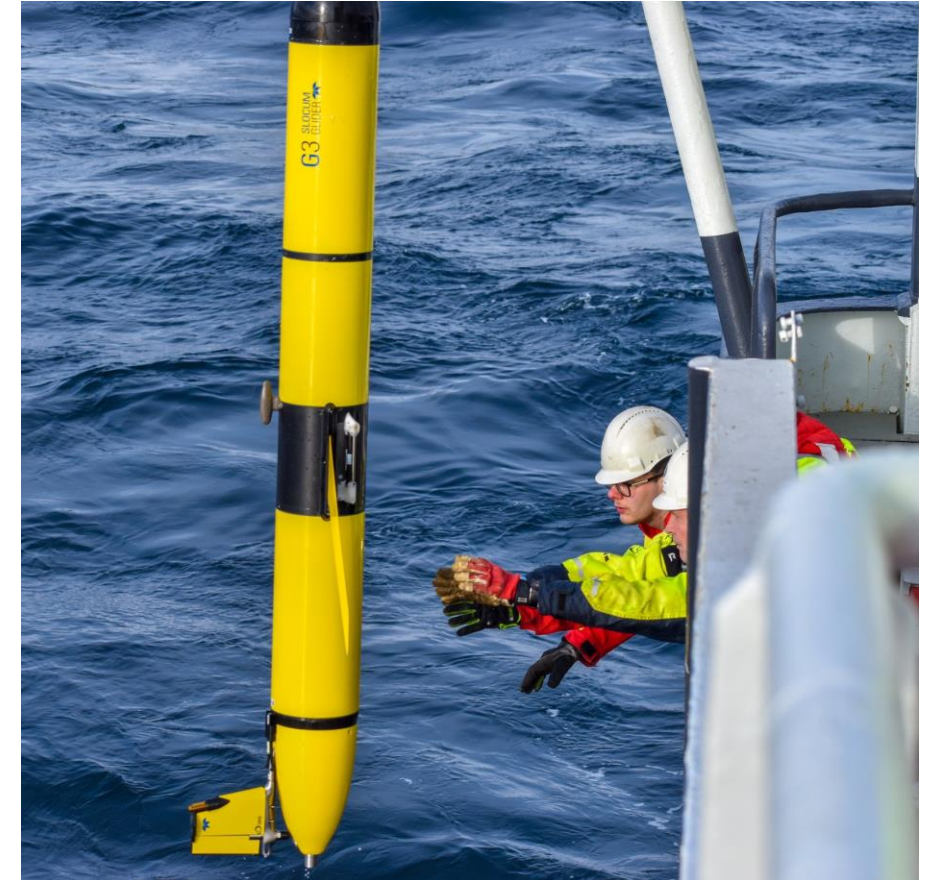
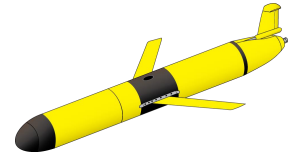


Photo: Fiona Elliot



# The Glider Aborted – *now what?*



- **Send the following commands to the glider:**
  - `why?` (explains why glider aborted)
  - `use` (prints list of installed devices)
  - `where` (prints surface dialog to screen) (callback 2 0?)
- **Download the relevant MLG (maybe the DBD)**
- **Increase time in GliderDOS so that glider doesn't sequence into `lastgasp.mi`**
- **Run callback 30 script**
  - Remember: GPS & Iridium share the same antenna. Only one works at a time!

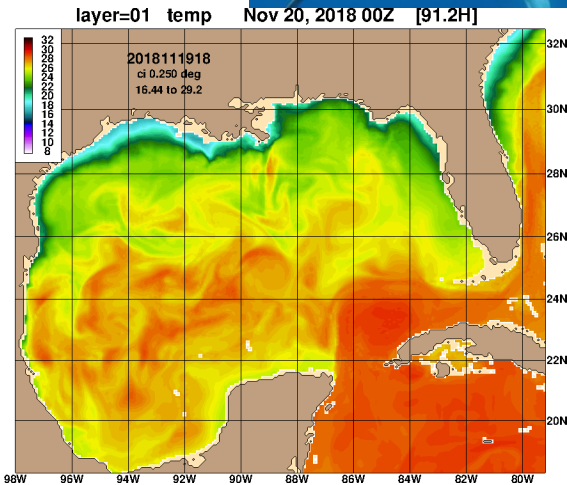
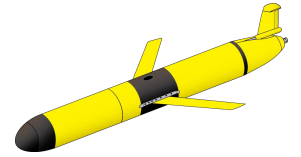
<https://datahost.webbresearch.com/files.php?cwd=/glider/production/doco/how-it-works/abort-sequences.txt>

<https://datahost.webbresearch.com/viewforum.php?f=7>



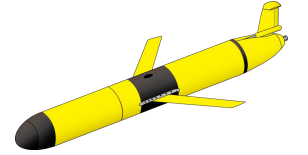


# Dangers to Gliders



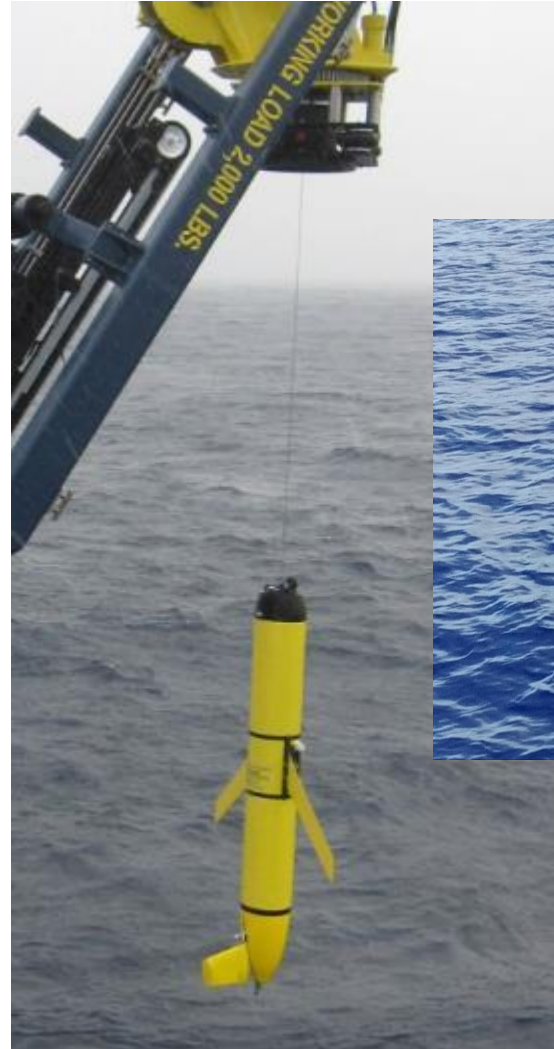
- People – strangers and technicians
- Shipping (transporting)
- Predators ([Remus Great White Shark video](#))
- Seals (green plug)
- Remora
- Low density surface layer (warm and fresh H2O)
- Trucks

# Recovery

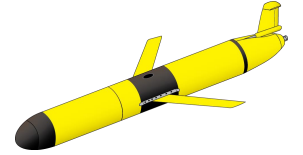


- Small boat?
- Big boat?
- Strobe on?
- Recovery system?
- GPS format on recovery boat: is translation needed?
- ARGOS data?

**MAKE A PLAN!**



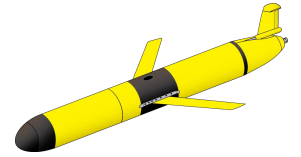
# Mission Writing



- **Start simple!**
- **Modify astock.mi**
  - Default mission
  - flies around 4 points in Ashumet Pond – where Teledyne Webb Research regularly flies and tests gliders
- **You will likely need to adjust:**
  - d\_target\_depth
  - max\_wpt\_distance
  - Surface intervals
    - No comms timeout
    - Every x minutes
    - Surfacing at every waypoint



# Mission Writing



## Modify Waypoints

- The `goto l10.ma` file controls waypoints

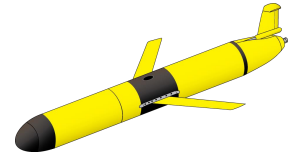
```
<start:b_arg>
b_arg: num_legs_to_run(nodim) -1 # loop
b_arg: start_when(enum) 0 # BAW_IMMEDIATELY
b_arg: list_stop_when(enum) 7 # BAW_WHEN_WPT_DIST
b_arg: initial_wpt(enum) -2 # closest
b_arg: num_waypoints(nodim) 4
<end:b_arg>
<start:waypoints>
-7032.0640 4138.1060
-7031.9200 4138.1090
-7031.9170 4138.0000
-7032.0610 4137.9980
<end:waypoints>
```



## Modify Dive Characteristics

- The `yo14.ma` file controls:
  - Dive to depth
    - must be less than max working depth
  - Climb to depth
    - must be shallower than 30m for piston-driven gliders
    - Inflecting at depths deeper than 30m will cause mission to abort (preventing damage to hardware)
  - Altitude
    - 6m off bottom for piston-driven gliders
    - 15m off bottom for oil-driven gliders

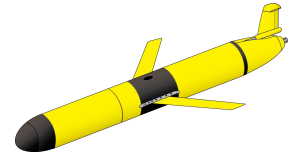
# Mission Writing



## Surface Behaviors – reasons to come to the surface

- No Comms
- Pitch Idle (pitch not commanded)
- Stack Idle (nothing commanded)
- Heading Idle
- Hit a Waypoint
- Every X seconds
- For UTC time
- When requested by science (most science sensors do not have this control)

# Science Sensors



## The “clothesline”

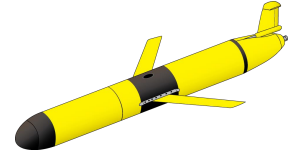
- Communication between the Flight & Science processors

## Specify sampling scheme in sampleXX.ma files

- state\_to\_sample (diving, climbing, hovering, on sfc)
- intersample\_time (# seconds between measurements)
- nth\_yo\_to\_sample (sample only  $n^{\text{th}}$  yo after  $1^{\text{st}}$  yo)
- intersample\_depth (# meters between measurements)
- min\_depth
- max\_depth

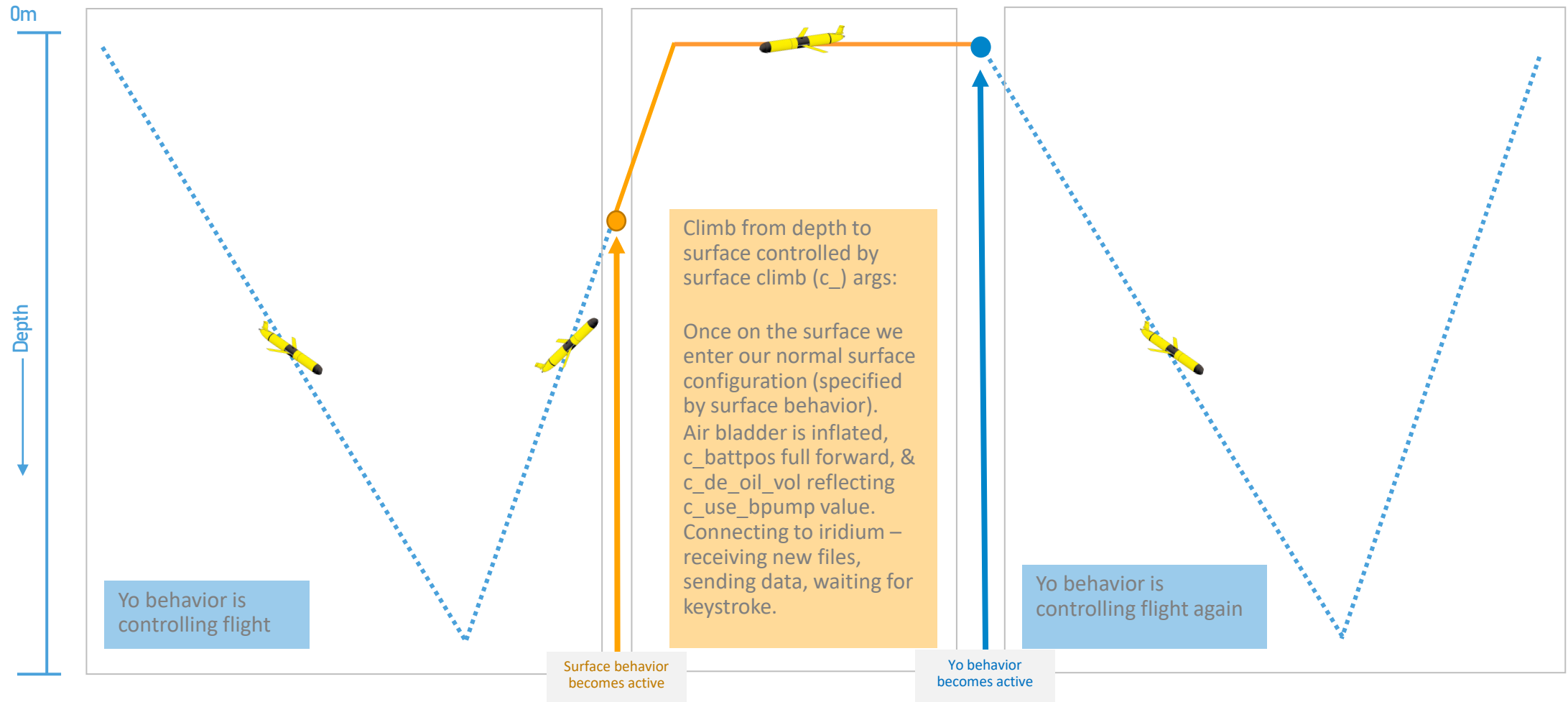
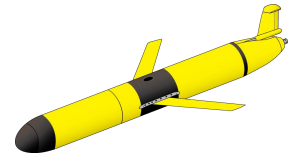




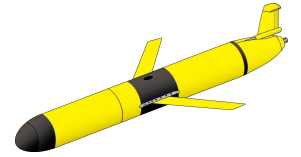


- **astock.mi Behaviors:**

- abend (abort behaviors)
- surface (surface behavior, surfac21.ma)
- surface (surface behavior, surfac22.ma)
- surface (surface behavior, surfac23.ma)
- surface (surface behavior, surfac24.ma)
- surface (surface behavior, surfac25.ma)
- surface (surface behavior, surfac26.ma)
- goto (waypoints, goto\_l10.ma)
- yo (dive and climb behavior, yo14.ma)
- sample (science control, sample10.ma)
- prepare\_to\_dive (needs to be here, but don't change)
- sensors\_in (needs to be here, but don't change)



# Scripts – *glider autopilot*



- SFMC is TWR recommended
- sends all \*.sbd and all \*.tbd
  - S \*.sbd \*.tbd -f irid -num=3
- dockzr all \*.ma files
  - !dockzr \*.\* -f irid
- Ctrl-F – ReRead .ma files
- Ctrl-W – Warning report (prints device list)
- Ctrl-R (dive!)

**Slocum Fleet Mission Control** 2021-09-05 17:13:52

Mission Control Configuration History Tools Admin About

Dock Server Scripts

### Factory Scripts

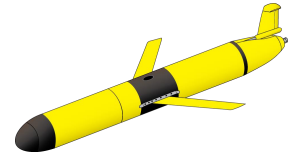
| Script Name          | In Use |
|----------------------|--------|
| IridCallback.xml     | Yes    |
| callbackPrimary.xml  | Yes    |
| glmpc-archive.xml    | No     |
| glmpc-direct-all.xml | No     |
| glmpc-direct.xml     | No     |
| resume.xml           | Yes    |
| sfmc-g3s-comp.xml    | No     |
| sfmc-g3s-num=3.xml   | No     |

### User Scripts

For group default:

| Script Name        | In Use | Delete |
|--------------------|--------|--------|
| Iridium            | No     |        |
| callback15.xml     | No     |        |
| callback20.xml     | No     |        |
| callback30.xml     | No     |        |
| callback30_mcw.xml | Yes    |        |
| callback5.xml      | No     |        |
| sfmc-g3s-3.xml     | No     |        |

# DBD format – *process glider data files*



- File naming convention

- File name on glider, example :

03100016

mission # on glider  
segment #

- Full file name, example :

capx594-2021-195-1-16

glider name  
year  
year day  
mission # on that day  
segment #

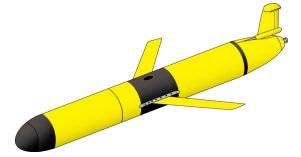
- Cache files

- Needed to process data files

- Processing

- SFMC (processed automatically)
- Executables (windoze-bin or linux-bin)

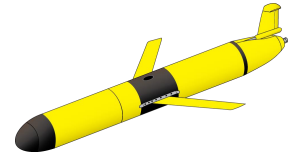
# Simulations – *practice before deployment*



- **Three different types of simulator are available**
  - Pocket Simulator (just a persistor or STM32 processor, all electronics and motors are simulated)
  - Shoebox Simulator (persistor or STMS32 processor & mainboard, all motors are simulated; can be connected to a science board)
  - Full glider (all motors & electronics are used, some sensors simulated)
- **simul.sim file controls the type of simulation**
- **This file must be placed in the flight config directory with the appropriate text:**
  - no\_electronics (pocket simulator)
  - just\_electronics (shoebox simulator)
  - on\_bench (full glider)



# Simulations — *putting the glider into simulation mode*



simul?

*# verify the glider is not currently in simulation*

To create the simulation file:

capture config/simul.sim

on\_bench

ctrl-c

dir config

*# verify simul.sim file is in config directory*

Type config/simul.sim

*# verify the contents of the simulation file are correct*

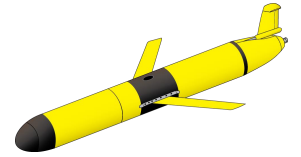
Exit reset

*# to force the glider to read the simulation file*

Simul?

*# verify the glider is in simulation mode*

# Simulations – *taking the glider out of simulation mode*



simul?

*# verify the glider is currently in simulation*

Remove the simulation file:

del config/simul.sim

dir config

*# verify simul.sim file is deleted from config directory*

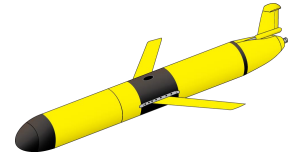
Exit reset

*# to force the glider to read the simulation file*

Simul?

*# verify the glider is NOT in simulation mode*

# Simulations – *loading simulation variables*



- loadsim.mi

- Set your variables in this file (sensors with prefix “s\_”)

- loadmission loadsim.mi

```
sensor: s_ini_lat(deg) 4138.1060      # Ashumet pond
sensor: s_ini_lon(deg) -7032.0640      # Ashumet pond

sensor: s_water_depth_avg(m)          200.0 #master data default is 30
sensor: s_water_depth_delta(m)        0.0
sensor: s_water_depth_wavelength(m)   100.0

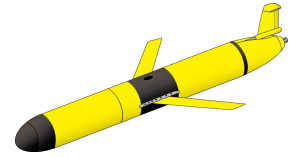
sensor: s_wind_speed(m/s)             9.0   # how fast the wind is blowing,
                                           # 3.0 ==> 5.4 knots
sensor: s_wind_direction(rad) 0.0        # Direction wind is blowing FROM

sensor: s_water_speed(m/s)             0.05  # Current speed, 0.5 ==> 1knot
sensor: s_water_direction(rad) 4.712      # direction current is going TO,
                                           # toward the west
```

- Tips for Simulation:

- You can modify autoexec.mi for simulation
- Change the name? (unit\_XXX\_sim)
- Uninstall iridium?
- For pocket simulators, set the time & date

# Surface Drift

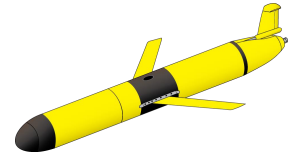


```
> put u_max_time_in_gliderdos 3600  
> callback 30
```

Assign a callback-30 script on the iridium server

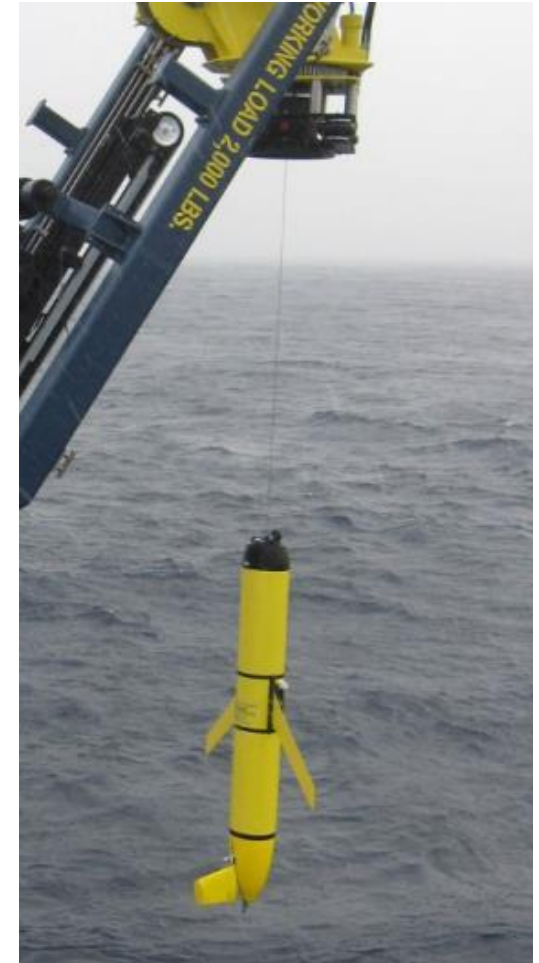
# Nose Recovery System

— *this will cause the glider to burn the nose release!!!*

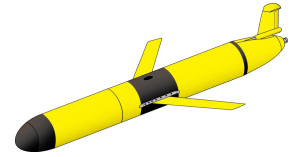


The command to burn the nose release wire to recover using the nose recovery system:

```
put c_recovery_on 1
```



# Notes - Regarding Training & Qualified Personnel

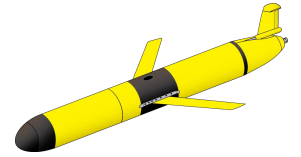


- **Only** trained and qualified personnel should operate and maintain the glider.
- Teledyne Webb Research conducts regular training sessions several times a year. Glider users should attend a training session and understand basic glider concepts and terminology.
- Contact [glidersupport@teledyne.com](mailto:glidersupport@teledyne.com) for information regarding training sessions.
- Company policy is to fully support only properly trained individuals and groups.
- Only personnel who have attended a Teledyne Webb Research training session should use this document





# Notes - *Regarding Training & Qualified Personnel*

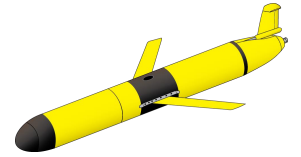


## Continued & Advanced Training

- Come back to another session in a year or two as a refresher and bring your intermediary questions
  - 2-4 sessions a year
- Watch for information about Advanced Training sessions in forum and by email
  - 1 a year



# Thank You!



We would like to thank you for attending this training session and look forward to assisting you as you deploy your gliders!

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Photo: Ben Allsup

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